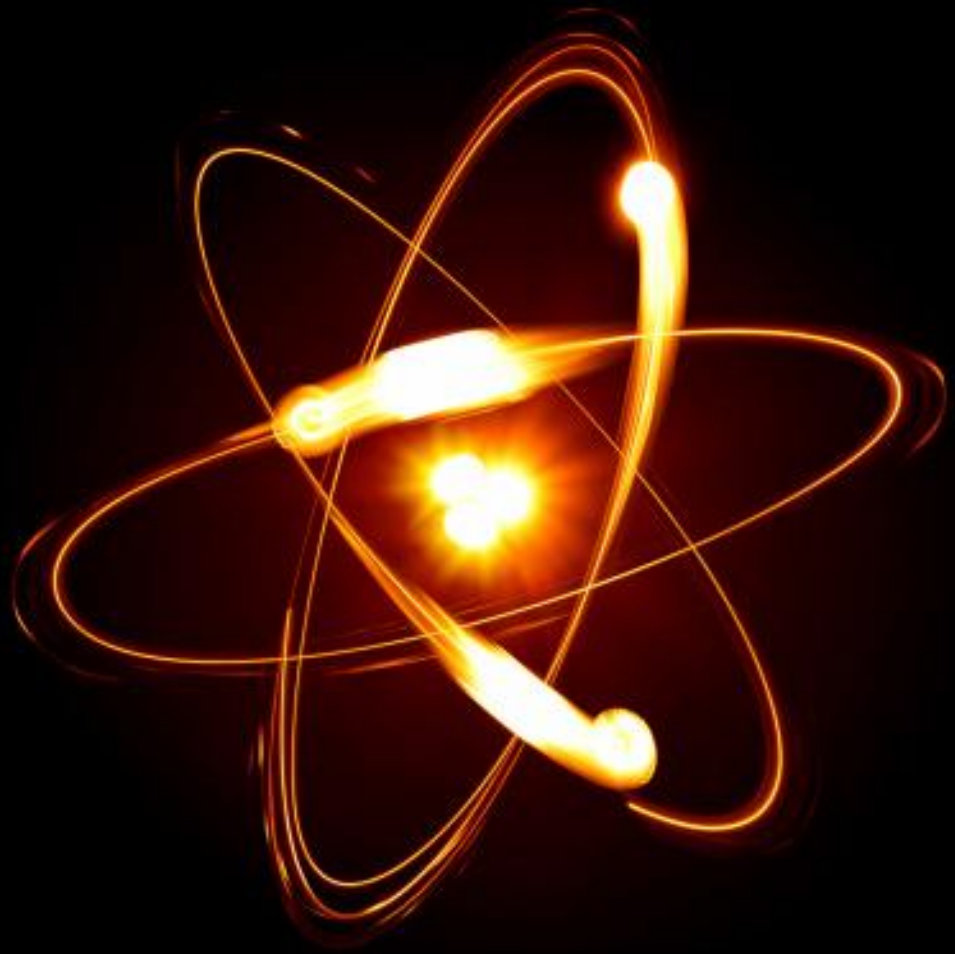
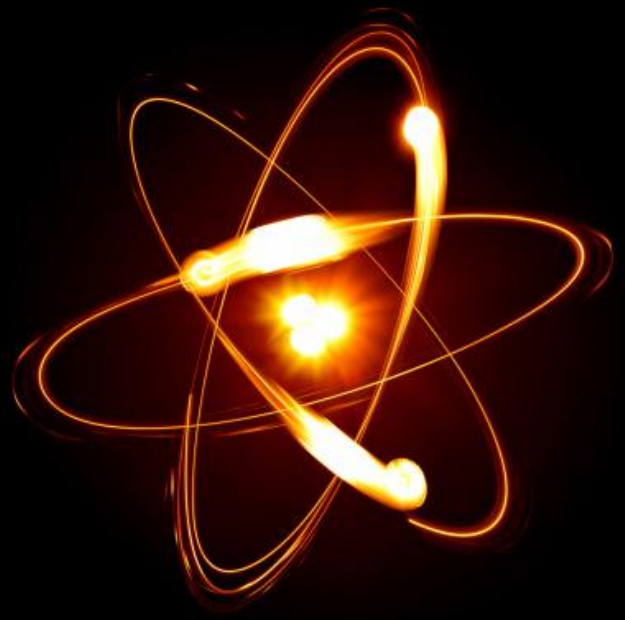
Chapter 7.

Solutions of the time-independent Schrödinger equation



Beatriz González del Río

Quantum Physics



Chapter 7.

Solutions of the time-independent Schrödinger equation

- The Zero Potential
- The Infinite Square Well Potential
- The Square Well Potential
- The Step Potential: energy less than step height
- The Step Potential: energy greater than step height
- The Barrier Potential
- Examples of barrier penetration
- 1-D Simple Harmonic Oscillator Potential

The Zero Potential

A particle of mass *m* and total energy E is moving in a region $V(x) = Cte$ (V(x)=0).

The particle is subjected to a force $F = -\frac{dV}{dx} = 0$

According to CM, the particle will be at rest OR move at constant velocity.

We have to solve the time-independent SE

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

$$\psi(x) = Ae^{ikx}$$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

And therefore

$$\Psi(x,t) = Ae^{i(kx-Et/\hbar)}$$

The Zero Potential

$$\Psi(x,t) = Ae^{i(kx-Et/\hbar)} = Ae^{i(kx-\omega t)} \qquad \omega = \frac{\hbar k^2}{2m}$$

$Ae^{i(kx-\omega t)}$ ⟶ Wave moving towards increasing *x*

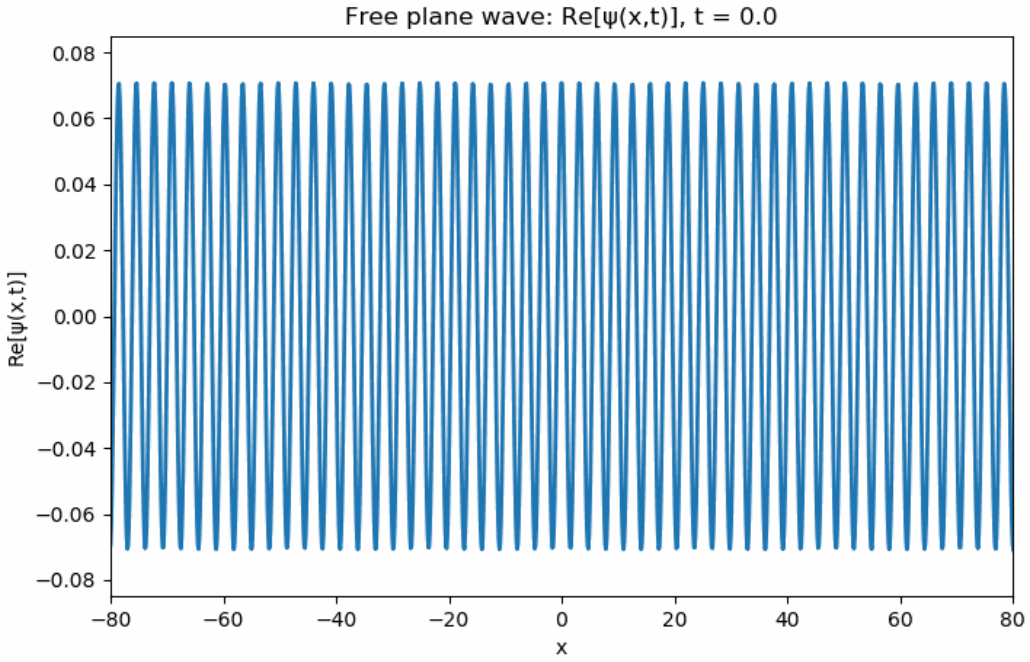
Real part $\cos(kx-\omega t)$

The zeros of the real part are at x_n such that

$$kx_n - \omega t = \left(n + \frac{1}{2}\right)\pi$$

Therefore $$x_n = \left(n + \frac{1}{2}\right)\pi/k + \omega t/k$$

But, there is another solution



The Zero Potential

Another solution

$$\psi(x) = Be^{-ikx} \qquad k = \frac{\sqrt{2mE}}{\hbar}$$

Therefore

$$\Psi(x,t) = Be^{-i(kx+Et/\hbar)} = Be^{-i(kx+\omega t)}$$

$Be^{-i(kx+\omega t)}$ $\longrightarrow$ Wave moving towards decreasing x .

The most general solution is

$$\psi(x) = Ae^{ikx} + Be^{-ikx}$$

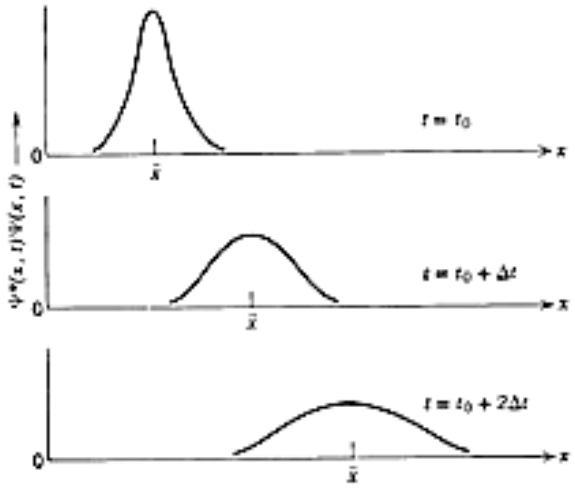


Figure 6-2 The probability density $\Psi^*\Psi$ for a group traveling wave function of a free particle. With increasing time the group moves in the direction of increasing x, and also spreads.

The Infinite Square Well Potential

1-D infinite square well potential

Describes a system of gas molecules or other particles which are strictly confined within a box of certain dimensions.

Particle in a box.

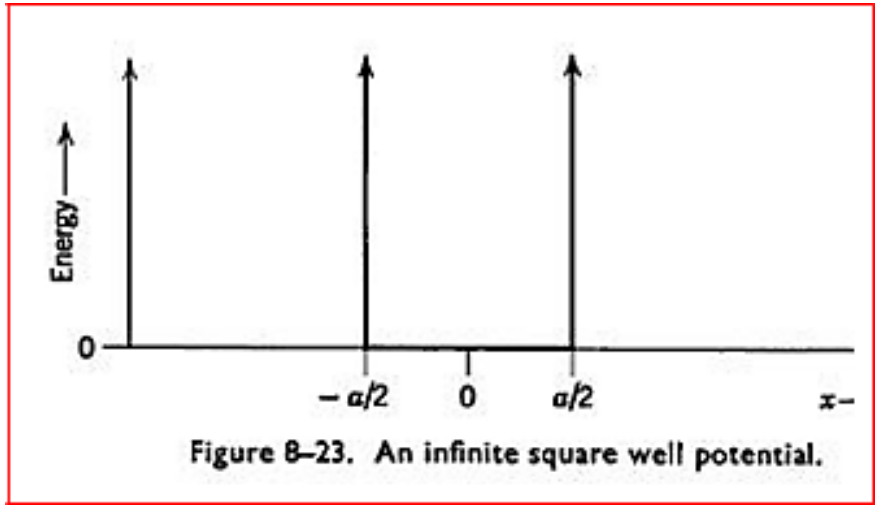


Figure 8–23. An infinite square well potential.

Binds a particle with any energy $E > 0$. In CM any energy is possible BUT in QM only certain discrete eigenvalues E_n are allowed.

There is only one region.

The Infinite Square Well Potential

Solve SE $$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+V(x)\psi(x)=E\psi(x)$$

$$\psi(x)=A'e^{ikx}+B'e^{-ikx}=A\sin kx+B\cos kx \qquad k=\frac{\sqrt{2mE}}{\hbar}$$

Boundary conditions (the derivative will be discontinuous at $x = \pm a/2$)

$$\psi(x=a/2)=0 \longrightarrow A\sin k(a/2)+B\cos k(a/2)=0$$

$$\psi(x=-a/2)=0 \longrightarrow -A\sin k(a/2)+B\cos k(a/2)=0$$

Addition and substraction give

$$2B\cos k\,a/2=0 \qquad 2A\sin k\,a/2=0$$

which cannot be satisfied simultaneously. (apart from A=0, and B=0)

The Infinite Square Well Potential

Two groups

- First class $A = 0$ and $\cos k(a/2) = 0$

  $\psi(x) = B\cos kx$ and $ka/2 = (2n+1)\pi/2$

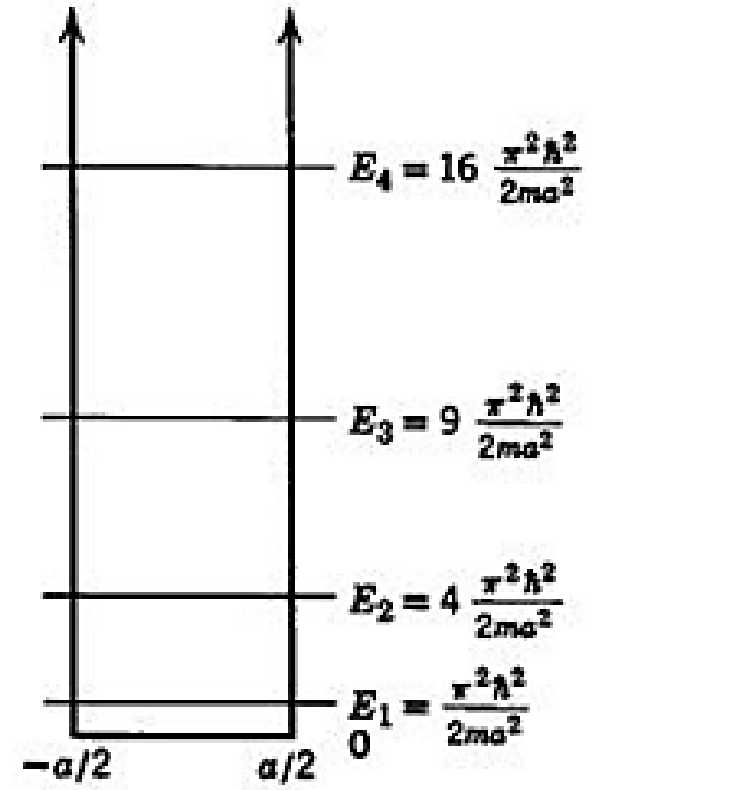
- Second class $B = 0$ and $\sin k(a/2) = 0$

  $\psi(x) = A\sin kx$ and $ka/2 = (2n)\pi/2$

More general

- First class $\psi_n(x) = B_n \cos k_n x$ and $k_n = n\,\pi/a$ $n = 1,3,5, \ldots$
- Second class $\psi_n(x) = A_n \sin k_n x$ and $k_n = n\pi/a$ $n = 2,4,6, \ldots$

The Infinite Square Well Potential



The first few eigenvalues for an infinite square

- From $k = \sqrt{2mE}/\hbar$ and $k_n = n\pi/a$

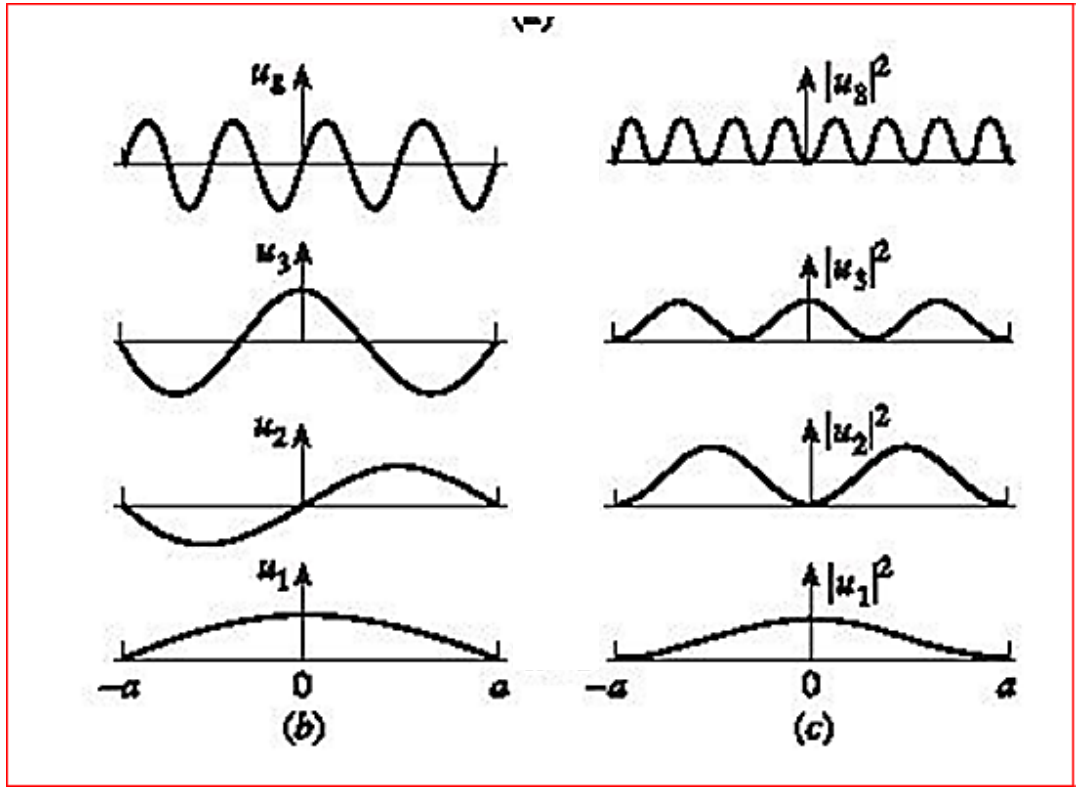
$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\pi^2\hbar^2 n^2}{2ma^2}$$

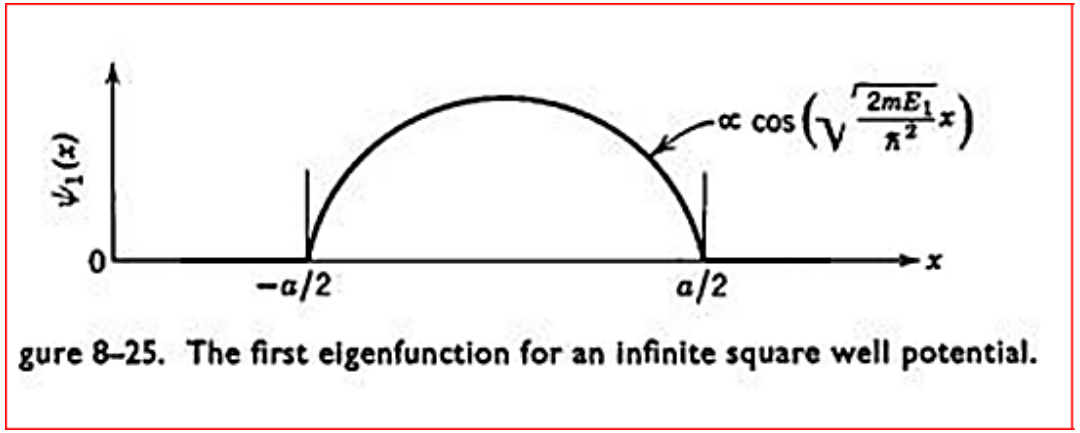
- Total energy is quantized

- First eigenvalue $E_1 = \frac{\pi^2\hbar^2}{2ma^2}$ zero-point energy.

  The particle cannot have zero energy

The Infinite Square Well Potential





gure 8–25. The first eigenfunction for an infinite square well potential.

The number of half-wavelengths is equal to its quantum number *n*

The number of nodes is *n* + 1

The Square Well Potential

Consider the following 1-D square well potential:

Used to describe situations in which a particle moves in a restricted region of space under the influence of forces which hold it in that region.

Example.- The potential acting on a conduction electron in a metal (depth ≈ 10 eV), a neutron in a nucleus (depth ≈ 50 MeV), ...

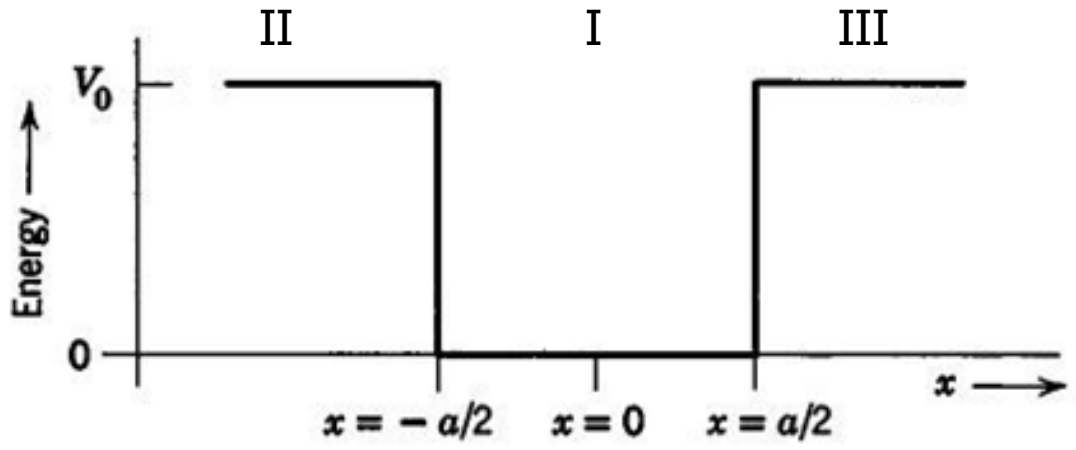
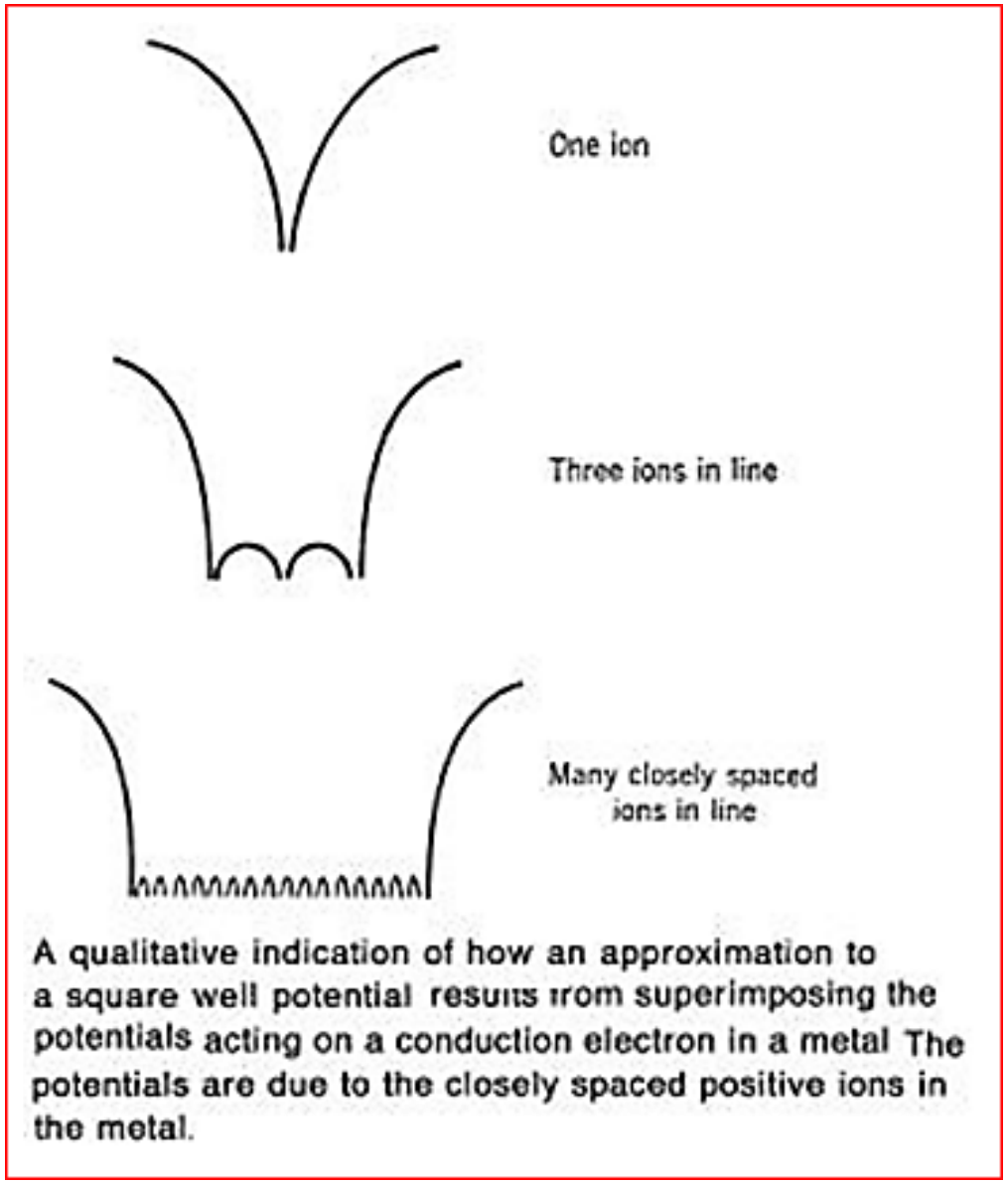


Figure 8–14. A square well potential.

The Square Well Potential



A qualitative indication of how an approximation to a square well potential results from superimposing the potentials acting on a conduction electron in a metal The potentials are due to the closely spaced positive ions in the metal.

The Square Well Potential

Case $E < V_0$

According to CM the particle would be bound to the region, bouncing back and forth. Any value of E is allowed.

According to QM, only **certain values of** E are allowed. This potential is used to represent those situations in which a particle moves in a restricted region of space.

Solve SE

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

There are three regions: I, II and III

The eigenfunction $\psi(x)$

$$\begin{cases} \psi_I(x) & -a/2 < x < a/2 \\ \psi_{II}(x) & x < -a/2 \\ \psi_{III}(x) & x > a/2 \end{cases}$$

The Square Well Potential

We solve SE in the different regions and we apply the conditions on the eigenfunction and its first derivative

We are interested in those solutions corresponding to energies $E < V_0$.

(The case $E > V_0$ is basically a situation with two step potentials.)

Regions II and III

$$\psi_{II}(x) = Ce^{k_2 x} + De^{-k_2 x}$$

$$\psi_{III}(x) = Fe^{k_2 x} + Ge^{-k_2 x}$$

$$k_2 = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

Region I

$$\psi_I(x) = Ae^{ik_1 x} + Be^{-ik_1 x} \equiv$$

$$\equiv A' \sin k_1 x + B' \cos k_1 x$$

$$k_1 = \frac{\sqrt{2mE}}{\hbar}$$

The Square Well Potential

- $\psi(x)$ is finite and single-valued for all x. $D = F = 0$

- $\psi(x)$ must be continuous for all x. We impose the condition at $x = -a/2$ and $x = a/2$

$$\psi_I(x = -a/2) = \psi_{II}(x = -a/2) \qquad \psi_I(x = a/2) = \psi_{III}(x = a/2)$$

- $\frac{d\psi(x)}{dx}$ must be continuous for all x. We impose the condition at $x = -a/2$ and $x = a/2$

$$\left.\frac{d\psi_I}{dx}\right|_{x=-a/2} = \left.\frac{d\psi_{II}}{dx}\right|_{x=-a/2} \qquad \left.\frac{d\psi_I}{dx}\right|_{x=a/2} = \left.\frac{d\psi_{III}}{dx}\right|_{x=a/2}$$

The Square Well Potential

$$\psi_I(x = -a/2) = \psi_{II}(x = -a/2)$$

$$\Rightarrow \quad Ce^{-k_2 a/2} = -A' \sin(k_1 a/2) + B' \cos(k_1 a/2) \tag{1}$$

$$\psi'_I(x = -a/2) = \psi'_{II}(x = -a/2)$$

$$\Rightarrow \quad Ck_2 e^{-k_2 a/2} = k_1 A' \cos(k_1 a/2) + k_1 B' \sin(k_1 a/2) \tag{2}$$

$$\psi_I(x = a/2) = \psi_{III}(x = a/2)$$

$$\Rightarrow \quad Ge^{-k_2 a/2} = A' \sin(k_1 a/2) + B' \cos(k_1 a/2) \tag{3}$$

$$\psi'_I(x = -a/2) = \psi'_{III}(x = -a/2)$$

$$\Rightarrow \quad -Gk_2 e^{-k_2 a/2} = k_1 A' \cos(k_1 a/2) - k_1 B' \sin(k_1 a/2) \tag{4}$$

The Square Well Potential

(3) - (1) $\Longrightarrow$ $(G - C)e^{-k_2 a/2} = 2A' \sin(k_1 a/2)$ (5)

(3) + (1) $\Longrightarrow$ $(G + C)e^{-k_2 a/2} = 2B' \cos(k_1 a/2)$ (6)

(2) - (4) $\Longrightarrow$ $(C + G)k_2 e^{-k_2 a/2} = 2B' k_1 \sin(k_1 a/2)$ (7)

(2) + (4) $\Longrightarrow$ $(C - G)k_2 e^{-k_2 a/2} = 2A' k_1 \cos(k_1 a/2)$ (8)

Four eqns and four constants ??????

The Square Well Potential

Dividing (7) by (6) we obtain:

$$k_1 \tan(k_1 a/2) = k_2 \qquad C + G \neq 0 \quad \text{and} \quad B' \neq 0 \tag{9}$$

Dividing (8) by (5) we obtain:

$$k_1 \frac{1}{\tan(k_1 a/2)} = -k_2 \qquad C - G \neq 0 \quad \text{and} \quad A' \neq 0 \tag{10}$$

These two equations can NOT be satisfied simultaneously

Assume the opposite. Then adding (9) and (10) we obtain:

$$k_1 \tan(k_1 a/2) + k_1 \frac{1}{\tan(k_1 a/2)} = 0 \tag{11}$$

Multiply by $\tan(k_1 a/2)$

$$k_1 \tan^2(k_1 a/2) + k_1 = 0 \qquad \Longrightarrow \qquad \tan^2(k_1 a/2) = -1$$

which is impossible. Therefore (9) and (10) can NOT be satisfied simultaneously

The Square Well Potential

Two possibilities

First class solutions

$$k_1 \tan(k_1\, a/2) = k_2 \qquad \text{and} \qquad A' = 0 \qquad \text{and} \qquad G - C = 0$$

From (6) we get

$$B' \cos(k_1\, a/2) = G e^{-k_2 a/2} \qquad\qquad G = B' \cos(k_1\, a/2) e^{k_2 a/2} = C$$

and the solutions are

$$\psi(x) \begin{cases} B' \cos(k_1 x) & -a/2 < x < a/2 \\ \left(B' \cos(k_1\, a/2) e^{k_2 a/2}\right) e^{k_2 x} & x < -a/2 \\ \left(B' \cos(k_1\, a/2) e^{k_2 a/2}\right) e^{-k_2 x} & x > a/2 \end{cases}$$

The Square Well Potential

Second class solutions

$$k_1 \frac{1}{\tan(k_1\, a/2)} = -k_2 \qquad B' = 0 \qquad G + C = 0$$

From (5) we get

$$A' \sin(k_1\, a/2) = G e^{-k_2 a/2} \qquad G = A' \sin(k_1\, a/2) e^{k_2 a/2} = -C$$

and the solutions are

$$\psi(x) \begin{cases} A' \sin(k_1 x) & -a/2 < x < a/2 \\ \left(-A' \sin(k_1\, a/2)\, e^{k_2 a/2}\right) e^{k_2 x} & x < -a/2 \\ \left(A' \sin(k_1\, a/2)\, e^{k_2 a/2}\right) e^{-k_2 x} & x > a/2 \end{cases}$$

The Square Well Potential

Consider again the first class solutions.

$k_1 \tan(k_1 a/2) = k_2$ and substitute for k_1 and k_2. Then

$$\sqrt{mEa^2/2\hbar^2}\tan\left(\sqrt{mEa^2/2\hbar^2}\right) = \sqrt{m(V_0 - E)a^2/2\hbar^2}$$

this eqn. must be solved for *E*.

Introduce the variable $\xi = \sqrt{mEa^2/2\hbar^2}$

$$p(\xi) \equiv \xi \tan\xi = \sqrt{mV_0a^2/2\hbar^2 - \xi^2} \equiv q(\xi)$$

The Square Well Potential

Graphical solution. Plot $p(\xi)$ and $q(\xi)$ as functions of ξ

$p(\xi)$ has zeros at $\xi = 0, \pi, 2\pi, \ldots, n\pi, \ldots$ and asymptotes at $\pi/2\,, 3\pi/2\,, 5\pi/2\,, \ldots$

$q(\xi)$ is a quarter circle of radius $\sqrt{mV_0a^2/2\hbar^2}$

The number of solutions depends on the radius of the quarter circle, (i.e. = 4)

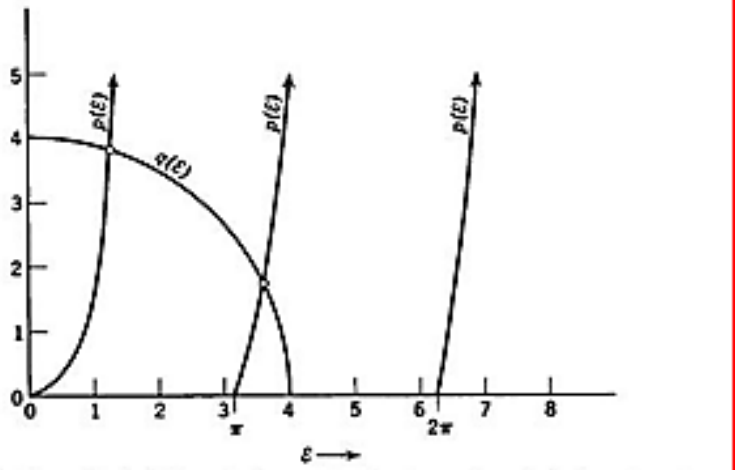


Figure 8–15. A graphical solution of the equation for eigenvalues of the first class of a particular square well potential. [Solution of $\varepsilon \tan \varepsilon = \sqrt{mV_0a^2/2\hbar^2 - \varepsilon^2}$ or $p(\epsilon) = q(\epsilon)$.]

Two solutions $\xi = 1.252$ and 3.595

$$E = \xi^2 \frac{2\hbar^2}{mV_0a^2} V_0 \approx \left(\frac{1.252}{4}\right)^2 V_0 = 0.0980\, V_0$$

$$E = \xi^2 \frac{2\hbar^2}{mV_0a^2} V_0 \approx \left(\frac{3.595}{4}\right)^2 V_0 = 0.808\, V_0$$

The Square Well Potential

Consider again the second class solutions.

$k_1 \frac{1}{\tan(k_1 a/2)} = -k_2$ and substitute for k_1 and k_2 .

In terms of the variable $\xi = \sqrt{mEa^2/2\hbar^2}$

$$r(\xi) \equiv -\xi \frac{1}{\tan \xi} = \sqrt{mV_0 a^2/2\hbar^2 - \xi^2} \equiv q(\xi)$$

The Square Well Potential

Graphical solution. Plot $r(\xi)$ and $q(\xi)$ as functions of ξ

$r(\xi)$ has zeros at $\xi = \pi/2\,, 3\pi/2\,, 5\pi/2\,, \ldots$ and asymptotes at $\pi, 2\pi, 3\pi, \ldots$

$q(\xi)$ is a quarter circle of radius $\sqrt{mV_0a^2/2\hbar^2}$

The number of solutions depends on the radius of the quarter circle

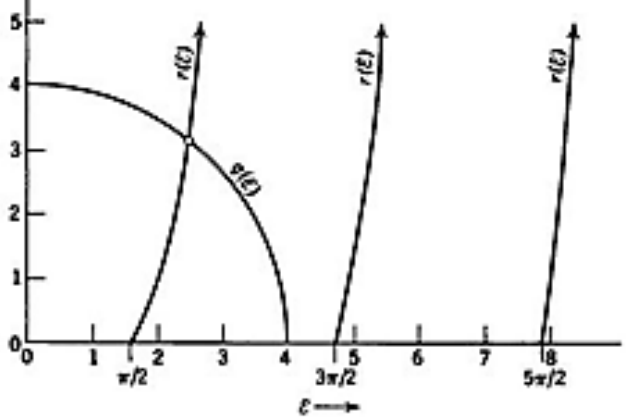


Figure 8-16. A graphical solution of the equation for eigenvalues of the second class of a particular square well potential. [Solution of $-\varepsilon \cot \varepsilon = \sqrt{mV_0a^2/2\hbar^2 - \varepsilon^2}$ or $r(\varepsilon) = q(\varepsilon)$.]

One solution

$\xi = 2.475$

$E = 0.383V_0$

The Square Well Potential

Eigenvalues corresponding to the previous three bound states

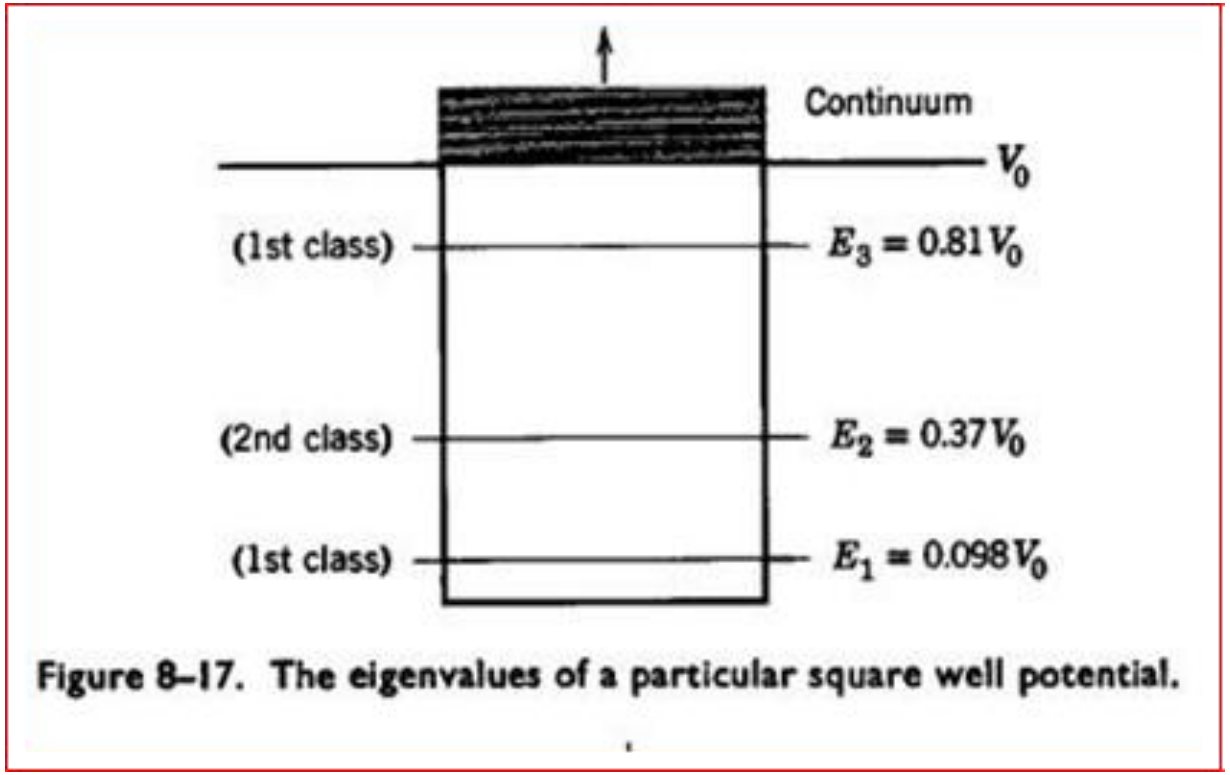


Figure 8–17. The eigenvalues of a particular square well potential.

For a very shallow/narrow potential there will be only a single level

With increasing values of $\sqrt{mV_0a^2/2\hbar^2}$ more levels will appear

The Square Well Potential

Eigenfunctions corresponding to the previous three bound states

$$\psi_1(x) = \begin{cases} 17.9\,\frac{1}{\sqrt{a}}\,e^{3.80\frac{x}{a/2}}, & x \leqslant -a/2 \\ 1.26\,\frac{1}{\sqrt{a}}\cos\left(1.25\,\frac{x}{a/2}\right), & -a/2 \leqslant x \leqslant a/2 \\ 17.9\,\frac{1}{\sqrt{a}}\,e^{-3.80\frac{x}{a/2}}, & x \geqslant a/2 \end{cases}$$

$$\psi_2(x) = \begin{cases} -18.6\,\frac{1}{\sqrt{a}}\,e^{3.16\frac{x}{a/2}}, & x \leqslant -a/2 \\ 1.23\,\frac{1}{\sqrt{a}}\sin\left(2.45\,\frac{x}{a/2}\right), & -a/2 \leqslant x \leqslant a/2 \\ 18.6\,\frac{1}{\sqrt{a}}\,e^{-3.16\frac{x}{a/2}}, & x \geqslant a/2 \end{cases}$$

$$\psi_3(x) = \begin{cases} -5.80\,\frac{1}{\sqrt{a}}\,e^{1.74\frac{x}{a/2}}, & x \leqslant -a/2 \\ 1.13\,\frac{1}{\sqrt{a}}\cos\left(3.60\,\frac{x}{a/2}\right), & -a/2 \leqslant x \leqslant a/2 \\ -5.80\,\frac{1}{\sqrt{a}}\,e^{-1.74\frac{x}{a/2}}, & x \geqslant a/2 \end{cases}$$

The Square Well Potential

Eigenfunctions corresponding to the previous three bound states

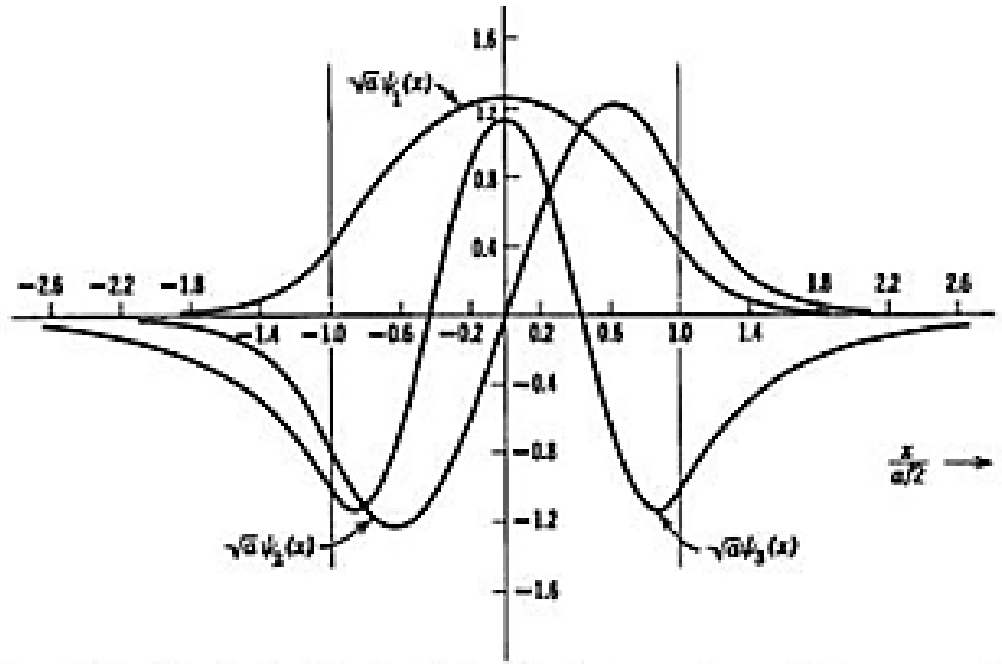


Figure 8–18. The eigenfunctions for the bound eigenstates of a particular square well potential.

Remember that $k_1 \propto E^{1/2}$ and $k_2 \propto (V_0 - E)^{1/2}$

The curvature increases with the $E \Rightarrow$ more oscillations $\Rightarrow$ greater k_1

The eigenfunctions extend into the region outside the well

Outside the well, when E increases, $V_0 - E$ decreases, k_2 decreases and we have a slower decay

The Square Well Potential

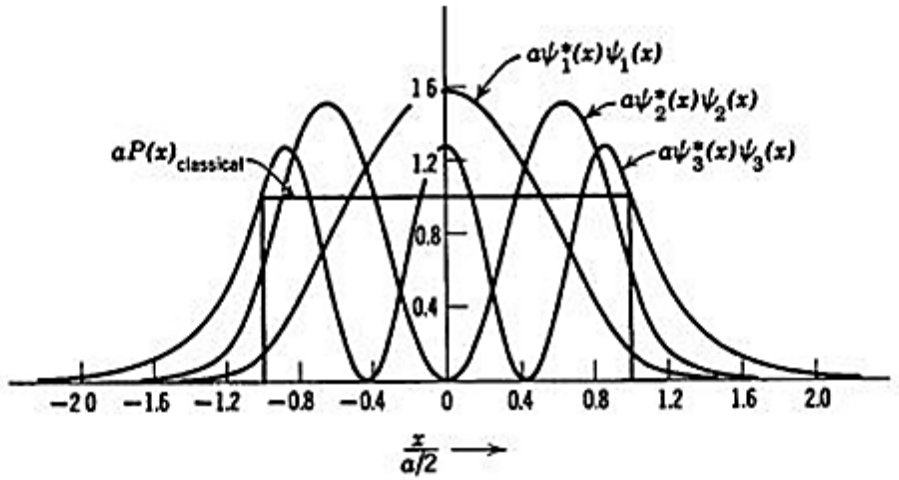


Figure 8–19. The probability densities for the bound eigenstates of a particular square well potential.

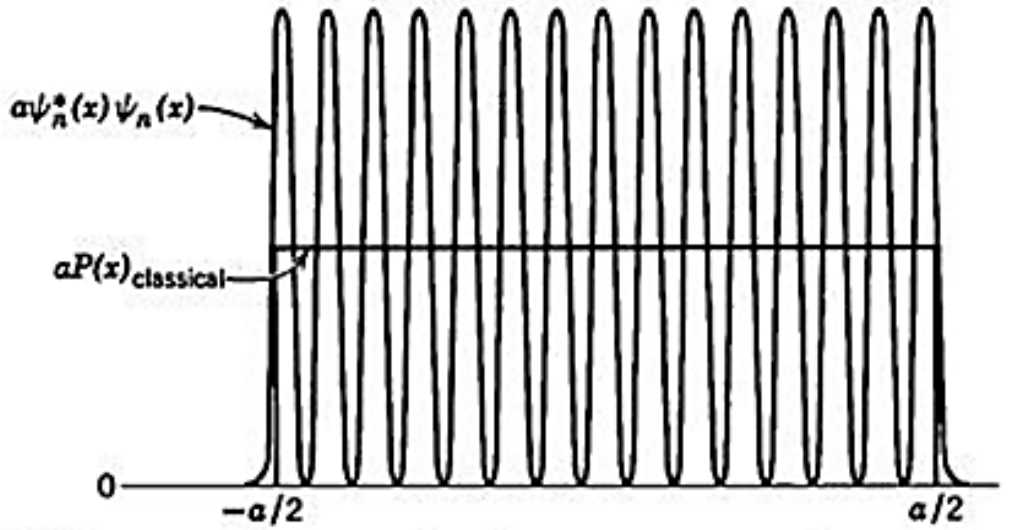


Figure 8–20. Illustrating the approach to the classical limit of the probability density for a square well potential.

The Square Well Potential

What happens to the eigenfunctions when the walls of the square well become very high?

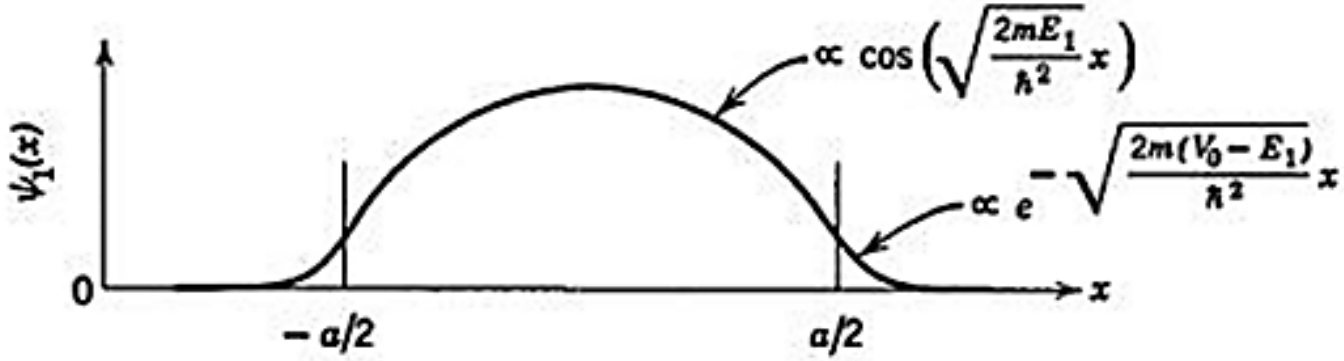
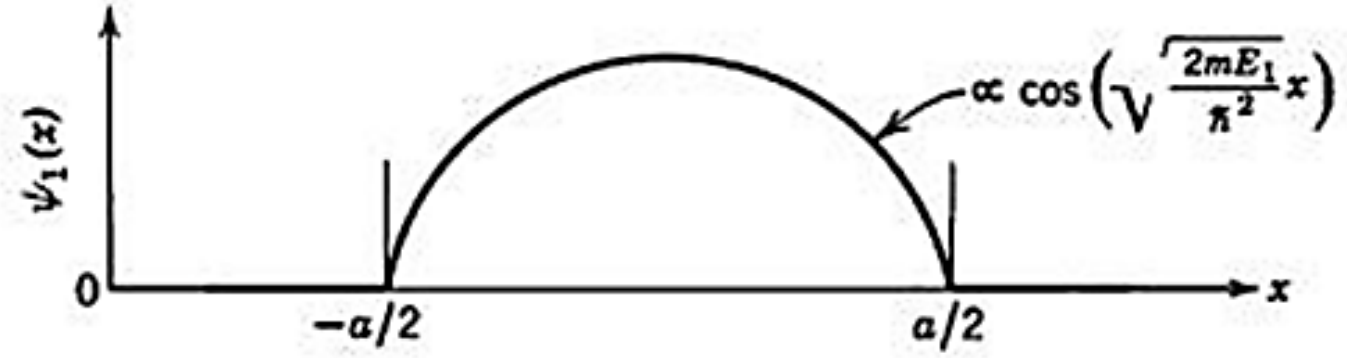


Figure 8–24. The first eigenfunction for a finite square well potential.

The Square Well Potential

When $V_0 \to \infty$, the eigenfunction becomes (infinite square well potential)

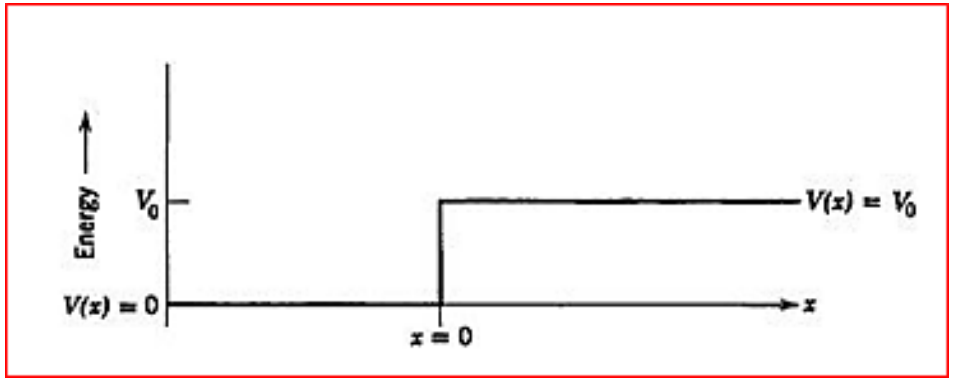


igure 8–25. The first eigenfunction for an infinite square well potential.

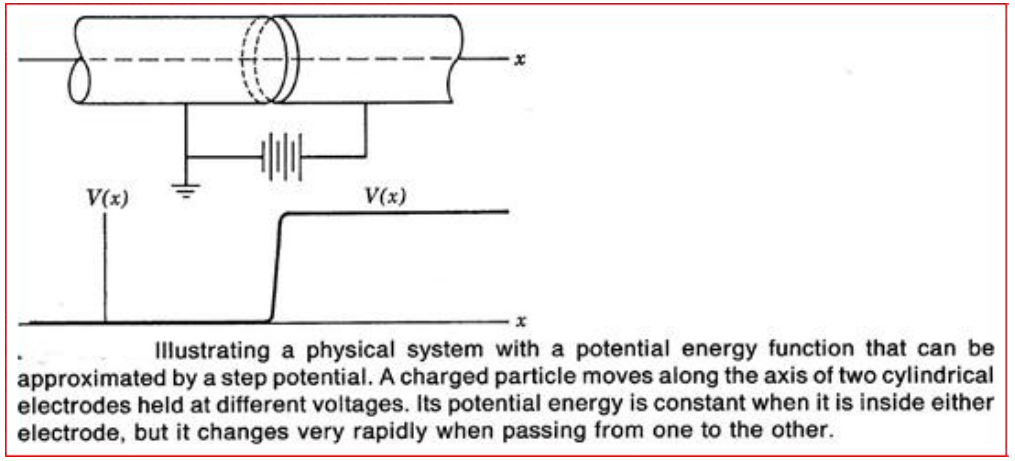
The Step Potential

Particle of mass *m* and total energy *E* is in the region $x < 0$ and it moves toward the point $x = 0$

- At $x = 0$ the particle is subjected to a force $F = -\frac{dV}{dx}$



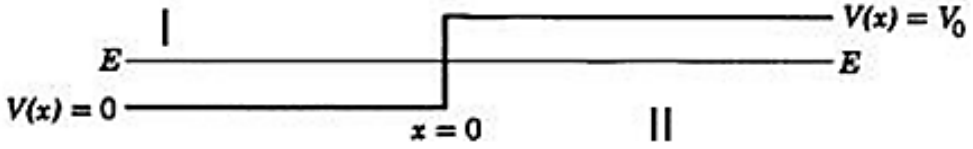
- The motion of the particle after experiencing the force depends, in CM, of the relation between *E* and V_0



Illustrating a physical system with a potential energy function that can be approximated by a step potential. A charged particle moves along the axis of two cylindrical electrodes held at different voltages. Its potential energy is constant when it is inside either electrode, but it changes very rapidly when passing from one to the other.

The Step Potential: Energy less than step height

Case when $E < V_0$



According to CM, the particle will reverse its motion at $x = 0$ and it will return with the same momentum magnitude

According to QM, we have to solve SE. There are two regions I and II

The eigenfunction

$$\psi(x) \begin{cases} \psi_I(x) & x < 0 \\ \psi_{II}(x) & x > 0 \end{cases}$$

The Step Potential: Energy less than step height

The time-independent SE adopts different forms according to the region

Region I

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_I(x)}{dx^2} = E\psi_I(x)$$

$$\psi_I(x) = Ae^{ik_1x} + Be^{-ik_1x}$$

$$k_1 = \frac{\sqrt{2mE}}{\hbar}$$

Region II

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_{II}(x)}{dx^2} + V_0\psi_{II}(x) = E\psi_{II}(x)$$

$$\psi_{II}(x) = Ce^{k_2x} + De^{-k_2x}$$

$$k_2 = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

There are four integration constants *A, B, C, D*. Apply conditions

a) $\psi(x)$ must be finite for all $x \to \infty$ Therefore $C = 0$

The Step Potential: Energy less than step height

b) $\psi(x)$ must be single-valued for all x

c1) $\psi(x)$ must be continuous for all x. We have to impose it at $x = 0$

$$\psi_I(x=0) = \psi_{II}(x=0) \longrightarrow A + B = D$$

c2) $\frac{d\psi(x)}{dx}$ must be continuous for all x. We have to impose it at $x = 0$

$$\left.\frac{d\psi_I}{dx}\right|_{x=0} = \left.\frac{d\psi_{II}}{dx}\right|_{x=0} \longrightarrow A - B = \frac{ik_2 D}{k_1}$$

Combining **c1)** and **c2)** we finally obtain

$$A = \frac{D}{2}\left(1 + \frac{ik_2}{k_1}\right) \qquad B = \frac{D}{2}\left(1 - \frac{ik_2}{k_1}\right)$$

The Step Potential: Energy less than step height

The eigenfunction will be

$$\psi(x) = \begin{cases} \frac{D}{2}\left(1+\frac{ik_2}{k_1}\right)e^{ik_1x} + \frac{D}{2}\left(1-\frac{ik_2}{k_1}\right)e^{-ik_1x} & x \le 0 \\ De^{-k_2x} & x \ge 0 \end{cases}$$

And the wave function $\Psi(x,t)$

$$\Psi(x,t) = \begin{cases} Ae^{i(k_1x-Et/\hbar)} + Be^{-i(k_1x+Et/\hbar)} & x \le 0 \\ De^{-k_2x}e^{-iEt/\hbar} & x \ge 0 \end{cases}$$

The Step Potential: Energy less than step height

Physical interpretation

$Ae^{i(k_1 x - Et/\hbar)}$ → Wave moving towards $x = 0$. Incident particle

$Be^{i(-k_1 x - Et/\hbar)}$ → Wave moving towards $x \to -\infty$. Reflected particle

Probability Current Density $J(x) = \frac{i\hbar}{2m}\left[\psi \frac{d\psi^*}{dx} - \psi^* \frac{d\psi}{dx}\right]$

Consider a wave $\psi(x) = Ae^{ikx}$. Then $J(x) = |A|^2 \frac{\hbar k}{m} = |A|^2 \mathrm{v}$

Take the eigenfunction for the step potential (two regions)

$J_I(x) = \frac{\hbar k_1}{m}\left(|A|^2 - |B|^2\right)$ $J_{II}(x) = 0$

Notice that $J_I(x) = J_i(x) - J_r(x)$

The Step Potential: Energy less than step height

The probability current densities are used to calculate the so-called reflection and transmission coefficients.

The reflection coefficient R is defined as $R = J_r/J_i$

Gives the probability that an incident particle be reflected

Obviously R + T = 1

The transmission coefficient T is defined as $T = J_t/J_i$

Gives the probability that an incident particle be transmitted

Apply to the step potential (energy less than step height)

$$R = \frac{J_r}{J_i} = \frac{|B|^2}{|A|^2} = 1$$

$$T = \frac{J_t}{J_i} = \frac{0}{|A|^2} = 0$$

The Step Potential: Energy less than step height

Notice that the solution can also be written as

$$\Psi(x,t) = \begin{cases} \left(D\cos k_1 x - D\dfrac{k_2}{k_1}\sin k_1 x\right)e^{-iEt/\hbar} & x \le 0 \\ De^{-k_2 x}e^{-iEt/\hbar} & x \ge 0 \end{cases}$$

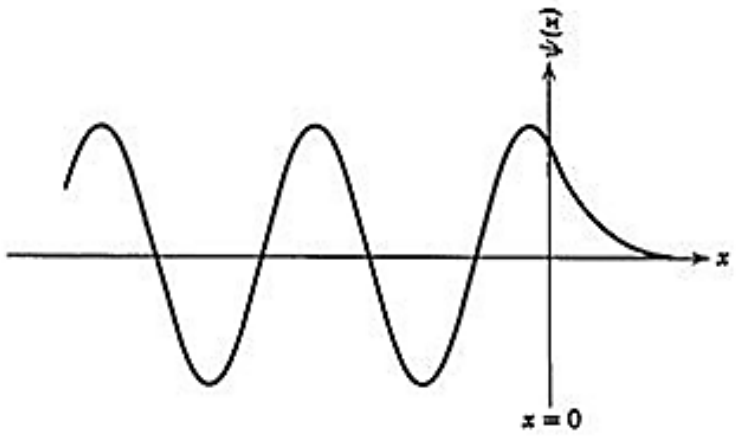


Figure 8–5. An eigenfunction for a step potential in the case in which classical mechanics predicts that the particle is reflected.

The Step Potential: Energy less than step height

Probability density in Region II. $\Psi^*\Psi = D^*De^{-2k_2x}$

Decreases quickly with increasing x but there is a finite probability of finding the particle in the region $x \geq 0$. This is absolutely forbidden in CM.

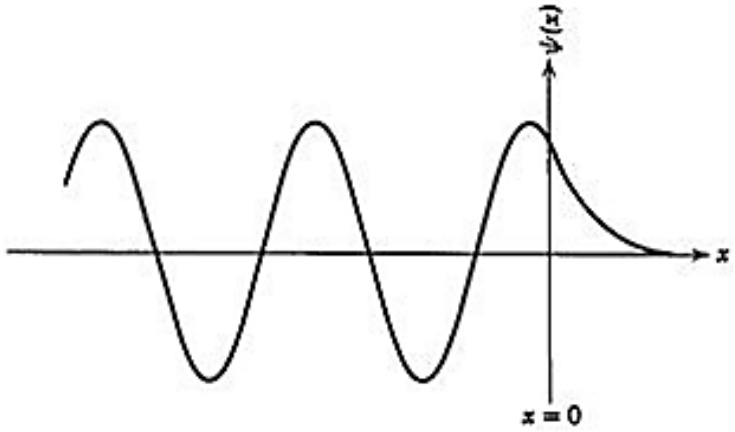


Figure 8–5. An eigenfunction for a step potential in the case in which classical mechanics predicts that the particle is reflected.

This is called the penetration of the classically excluded region. The penetration distance is defined as

$$\Delta I_x \approx \frac{1}{k_2} = \frac{\hbar}{\sqrt{2m(V_0 - E)}}$$

The Step Potential: Energy less than step height

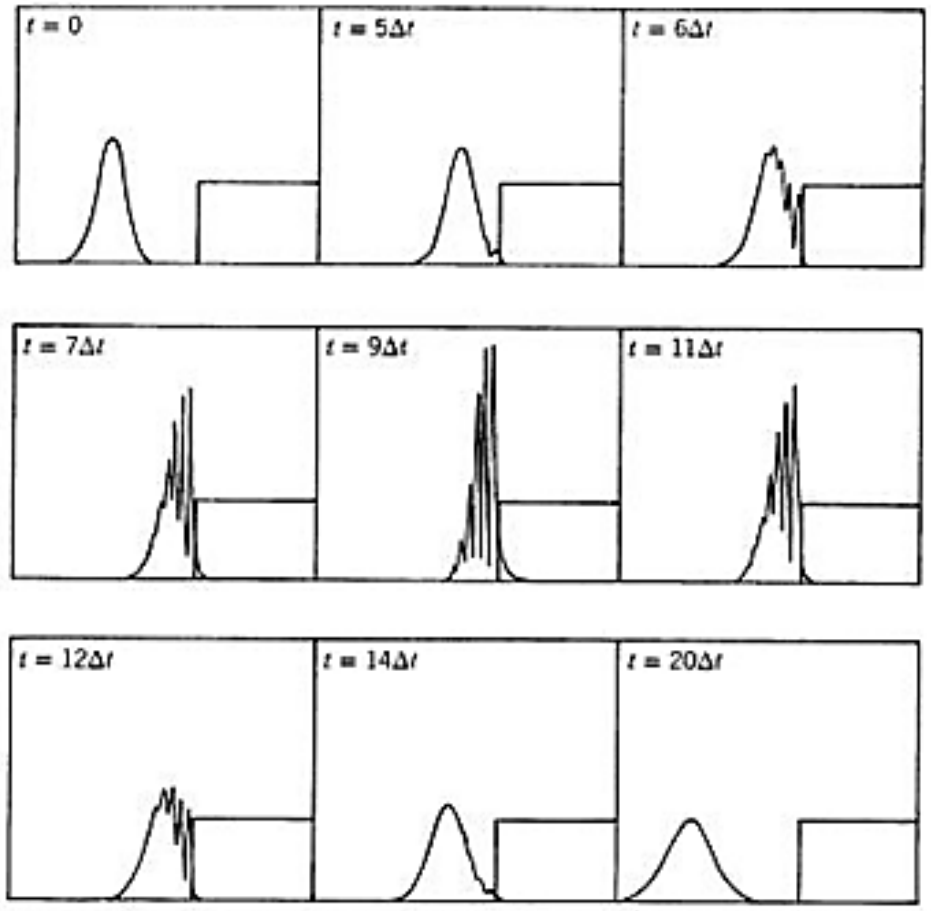
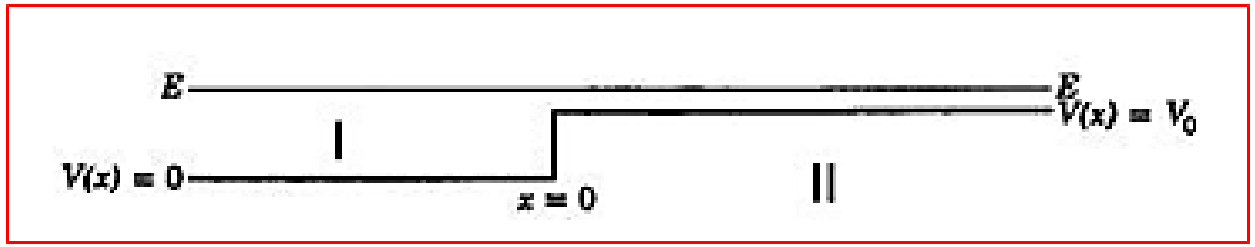


Figure 6-8 A potential step, and the probability density $\Psi^*\Psi$ for a group wave function describing a particle incident on the step with total energy less than the step height. As time evolves, the group moves up to the step, penetrates slightly into the classically excluded region, and then is completely reflected from the step. The complications of the mathematical treatment using a group are indicated by the complications of its structure during reflection.

The Step Potential: Energy greater than step height

Particle of mass m and total energy $E > V_0$ is in the region $x < 0$ and it moves toward the point $x = 0$

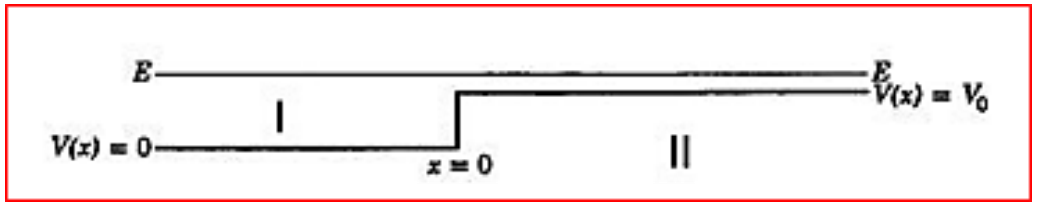


According to CM, the particle is moving with a momentum p_1 such that $E = p_1^2/2m$. When it reaches $x = 0$ the particle will continue moving in region II. Now, it has a momentum p_2 such that $E - V_0 = p_2^2/2m$

According to QM, the particle may be reflected, even when its energy $E > V_0$. This is a real effect, which can be observed in the photoelectric cells. When an electron absorbs a photon, if the energy of the photon is not much greater than the height of the potential, then the electron may be reflected back.

The Step Potential: Energy greater than step height

Solve the time-independent SE. There are two regions I and II



The eigenfunction $\psi(x)$

$$\psi(x) \begin{cases} \psi_I(x) & x < 0 \\ \psi_{II}(x) & x > 0 \end{cases}$$

Region I

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_I(x)}{dx^2} = E\psi_I(x)$$

$$\psi_I(x) = Ae^{ik_1x} + Be^{-ik_1x}$$

$$k_1 = \frac{\sqrt{2mE}}{\hbar}$$

Region II

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_{II}(x)}{dx^2} + V_0\psi_{II}(x) = E\psi_{II}(x)$$

$$\psi_{II}(x) = Ce^{ik_2x} + De^{-ik_2x}$$

$$k_2 = \frac{\sqrt{2m(E - V_0)}}{\hbar}$$

The second term in $\psi_{II}(x)$ describes a wave moving toward $x < 0$. There is no wave incident from $+\infty$ so we take D=0

The Step Potential: Energy greater than step height

a) $\psi(x)$ must be finite for all *x*

b) $\psi(x)$ must be single-valued for all *x*

c1) $\psi(x)$ must be continuous for all *x*. We have to impose it at *x* = 0

$$\psi_I(x=0) = \psi_{II}(x=0) \longrightarrow A + B = C$$

c2) $\frac{d\psi(x)}{dx}$ must be continuous for all *x*. We have to impose it at *x* = 0

$$\left.\frac{d\psi_I}{dx}\right|_{x=0} = \left.\frac{d\psi_{II}}{dx}\right|_{x=0} \longrightarrow A - B = \frac{k_2 C}{k_1}$$

Combining **c1)** and **c2)** we finally obtain

$$B = \frac{k_1 - k_2}{k_1 + k_2} A \qquad C = \frac{2k_1}{k_1 + k_2} A$$

The Step Potential: Energy greater than step height

The eigenfunction will be

$$\psi(x) = \begin{cases} Ae^{ik_1 x} + A\dfrac{k_1 - k_2}{k_1 + k_2} e^{-ik_1 x} & x \leq 0 \\ \dfrac{2k_1}{k_1 + k_2} Ae^{ik_2 x} & x \geq 0 \end{cases}$$

And the wave function $\Psi(x,t)$

$$\Psi(x,t) = \begin{cases} Ae^{i(k_1 x - Et/\hbar)} + Be^{-i(k_1 x - Et/\hbar)} & x \leq 0 \\ Ce^{ik_2 x} e^{-iEt/\hbar} & x \geq 0 \end{cases}$$

The Step Potential: Energy greater than step height

Compute the reflection and transmission coefficients

We take the eigenfunction for the step potential (two regions)

$$J_I(x) = \frac{\hbar k_1}{m}\left(|A|^2 - |B|^2\right) = \frac{\hbar k_1}{m}\left[|A|^2 - |A|^2\left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2\right] \equiv J_i - J_r$$

$$J_{II}(x) = \frac{\hbar k_2}{m}|C|^2 = \frac{\hbar k_2}{m}|A|^2\left(\frac{2k_1}{k_1 + k_2}\right)^2 \equiv J_t$$

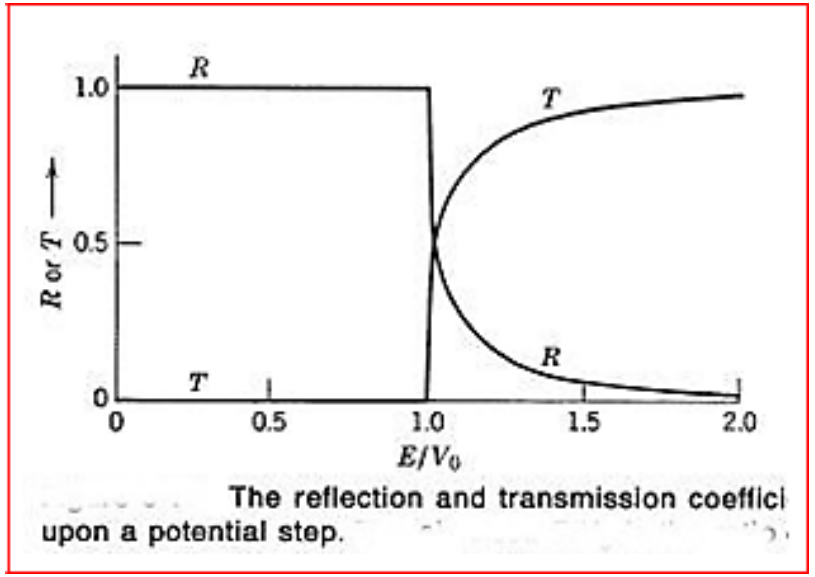
$$R = \frac{J_r}{J_i} = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2 \qquad T = \frac{J_t}{J_i} = \frac{k_2}{k_1}\left(\frac{2k_1}{k_1 + k_2}\right)^2 = \frac{4k_1k_2}{(k_1 + k_2)^2}$$

Notice that $R + T = 1$

The Step Potential: Energy greater than step height

- The incident probability current density is split into transmitted and reflected probability current densities.
- The particle does not vanish, it is either transmitted or is reflected
- The same results are obtained if the particle were incident upon the potential step in the direction of decreasing x from the region $x > 0$ (reciprocity property of waves in optics)
- By evaluating k_1 and k_2, it is found that

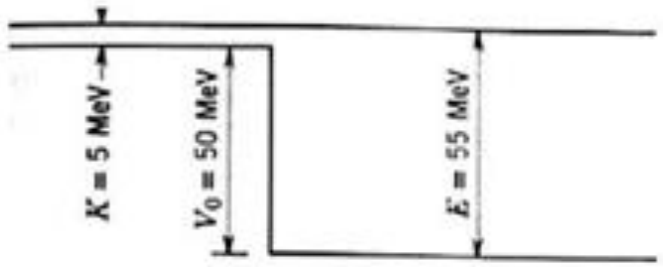
$$R = 1 - T \begin{cases} \left(\dfrac{1-\sqrt{1-V_0/E}}{1+\sqrt{1-V_0/E}}\right)^2 & E > V_0 \\ 1 & E < V_0 \end{cases}$$



The reflection and transmission coeffici
upon a potential step.

The Step Potential: Energy greater than step height

Example 6-3. When a neutron enters a nucleus, it experiences a potential energy which drops at the nuclear surface very rapidly from a constant external value V=0 to a constant internal value of about V=-50 Mev. The decrease in the potential is what makes it possible for a neutron to be bound in a nucleus. Consider a neutron incident upon a nucleus with an external kinetic energy K=5Mev, which is typical for a neutron that has just been emitted from a nuclear fission. Estimate the probability that the neutron will be reflected at the nuclear surface, thereby failing to enter and have its chance at inducing another nuclear fission.

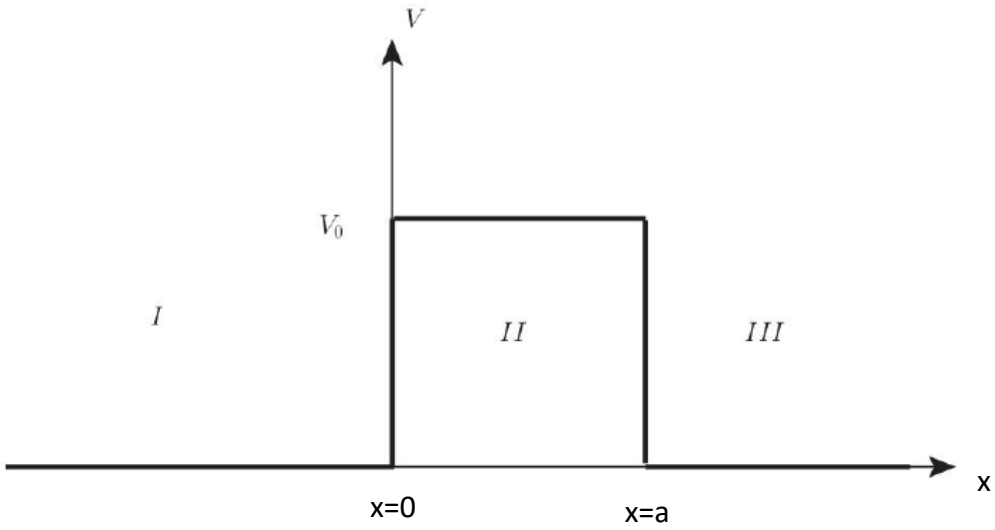


$$R = \left(\frac{1-\sqrt{1-\frac{50}{55}}}{1+\sqrt{1-\frac{50}{55}}}\right)^2 = 0.29$$

A neutron of external kinetic energy K incident upon a decreasing potential step of depth Vo, which approximates the potential it feels upon entering a nucleus. Its total energy, measured from the bottom of the step potential, is E.

The Barrier Potential

Particle of energy E in the region $x < 0$ is incident upon the barrier

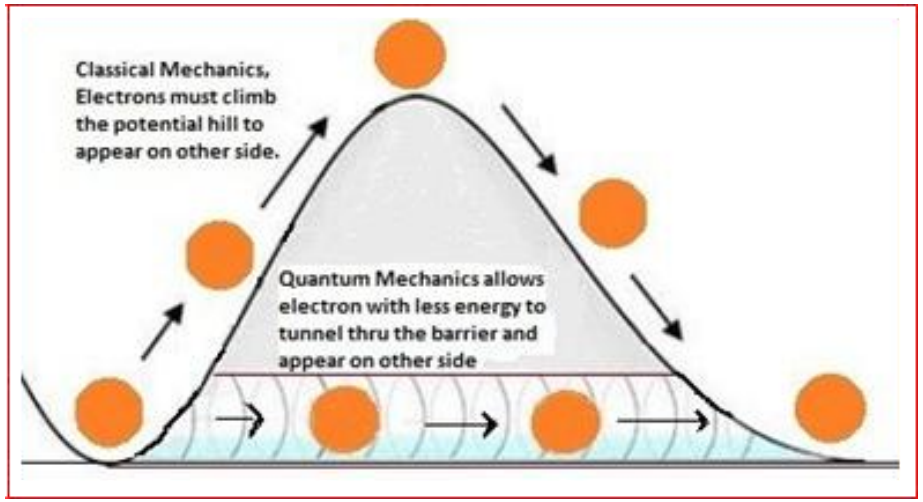


Classical Mechanics

- If $E > V_0$ the particle will be transmitted into the region $x > a$
- If $E < V_0$ the particle will be reflected

Quantum Physics

- If $E > V_0$ (although not much larger) there is some probability that the particle will be reflected
- If $E < V_0$ (although not much smaller) there is some probability that the particle will be transmitted



The Barrier Potential

There are three regions: I, II and III

The eigenfunction $\psi(x)$

$$\begin{cases} \psi_I(x) & x < 0 \\ \psi_{II}(x) & 0 < x < a \\ \psi_{III}(x) & x > a \end{cases}$$

Solutions for the time-independent SE

Regions I and III

Solutions for any $E > 0$

$$\psi_I(x) = Ae^{ik_1x} + Be^{-ik_1x}$$

$$\psi_{III}(x) = Ce^{ik_1x} + De^{-ik_1x}$$

$$k_1 = \frac{\sqrt{2mE}}{\hbar}$$

Region II

$E < V_0$

$$\psi_{II}(x) = Fe^{-k_2x} + Ge^{k_2x} \qquad k_2 = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

$E > V_0$

$$\psi_{II}(x) = F'e^{-ik'_2x} + G'e^{ik'_2x} \qquad k'_2 = \frac{\sqrt{2m(E - V_0)}}{\hbar}$$

The particle is incident from $x < 0 \Rightarrow$ D=0 (but cannot set G=0)

The Barrier Potential

Case $E < V_0$

- $\psi(x)$ is finite and single-valued for all *x*
- $\psi(x)$ must be continuous for all *x*. We impose it at *x* = 0 and *x* = *a*

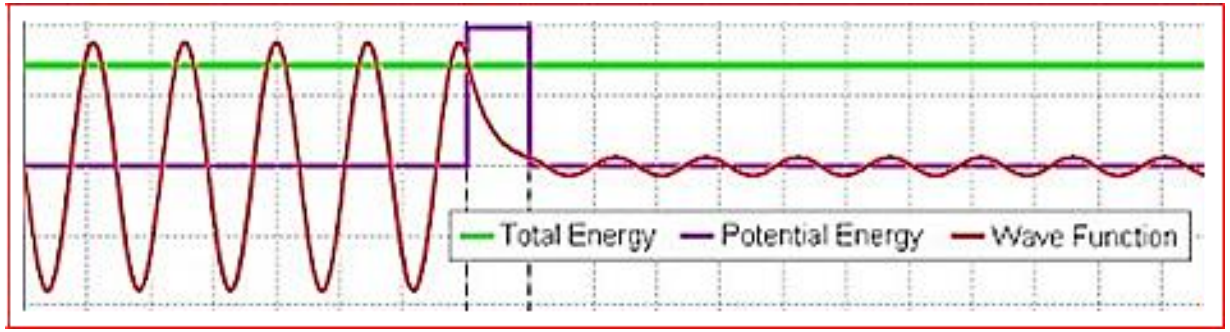
$$\psi_I(x=0) = \psi_{II}(x=0) \qquad \psi_{II}(x=a) = \psi_{III}(x=a)$$

- $\frac{d\psi(x)}{dx}$ must be continuous for all *x*. We impose it at *x* = 0 and *x* = *a*

$$\left.\frac{d\psi_I}{dx}\right|_{x=0} = \left.\frac{d\psi_{II}}{dx}\right|_{x=0} \qquad \left.\frac{d\psi_{II}}{dx}\right|_{x=a} = \left.\frac{d\psi_{III}}{dx}\right|_{x=a}$$

There are four equations which can be used to evaluate B, C, F and G in terms of A

The real part of a barrier penetration eigenfunction



The Barrier Potential

Transmission coefficient

$$T = \frac{J_t}{J_i} = \frac{\hbar k_1/m\,|C|^2}{\hbar k_1/m\,|A|^2} = \left[1 + \frac{(\sinh k_2 a)^2}{4(E/V_0)(1 - E/V_0)}\right]^{-1}$$

with $k_2 a = \sqrt{\frac{2mV_0 a^2}{\hbar^2}\left(1 - \frac{E}{V_0}\right)}$

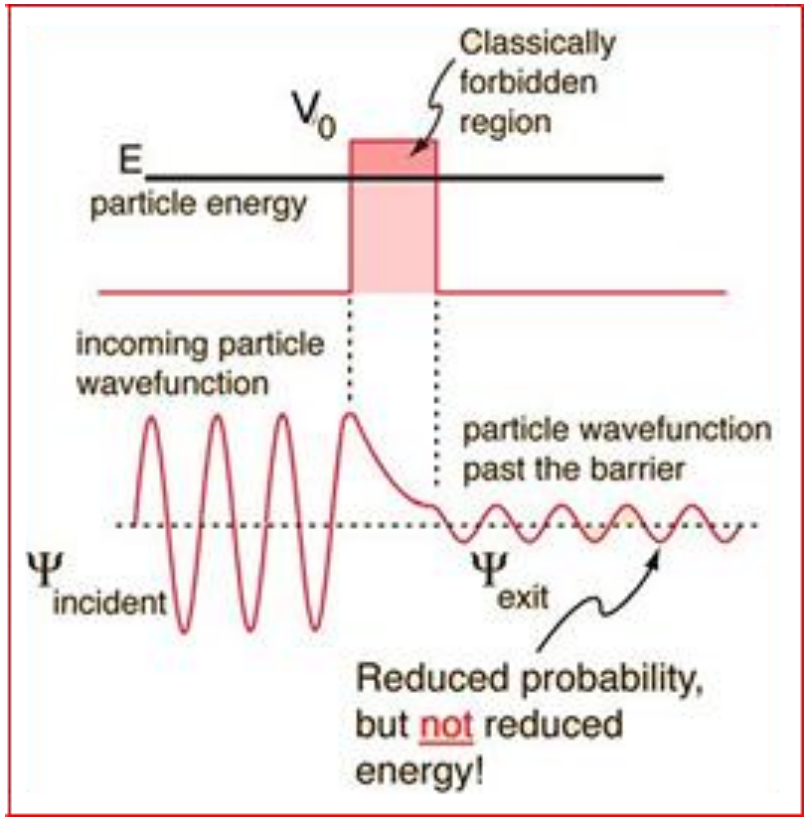
If $k_2 a \gg 1$

$$T \approx 16(E/V_0)(1 - E/V_0)e^{-2k_2 a}$$

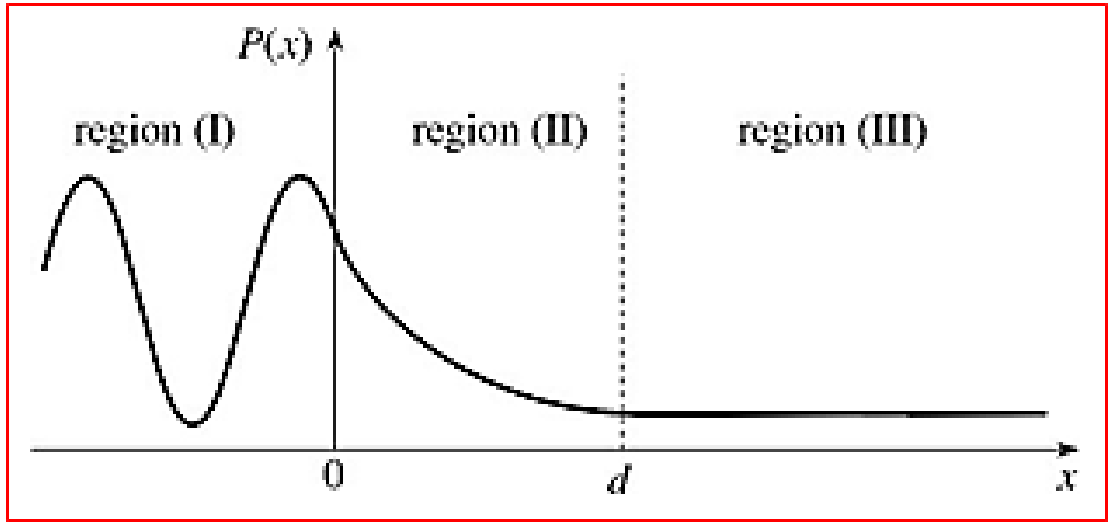
It predicts that a particle of mass *m* and total energy *E*, which is incident on a potential barrier of height $V_0 >$ *E* and finite thickness *a*, has a certain probability T of penetrating the barrier and appearing on the other side.

This is called **barrier penetration** and the particle is said to **tunnel** through the barrier.

The Barrier Potential



The real part of a barrier penetration eigenfunction



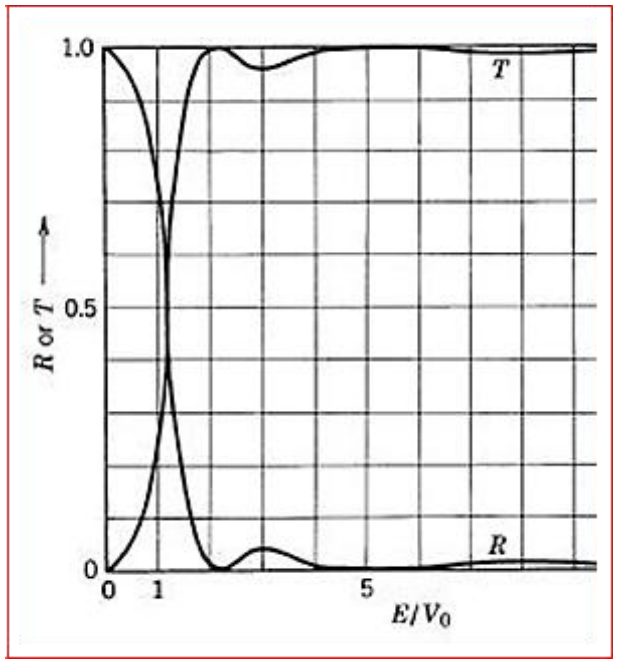
The probability density function for a barrier penetration situation

The Barrier Potential

Case $E > V_0$

$$T = \left[1 + \frac{(\sin k_2' a)^2}{4(E/V_0)(E/V_0 - 1)}\right]^{-1} \qquad \text{with } k_2' a = \sqrt{\frac{2mV_0 a^2}{\hbar^2}\left(\frac{E}{V_0} - 1\right)}$$

Example. An electron incident upon a rectangular barrier with V_0 = 10 eV and thickness a = 1.8 Å



- $2mV_0a^2/\hbar^2 \approx 9$
- T very small when $E/V_0 << 1$
- T ≈ 0.05 when $E/V_0 = 0.5$
- For $E/V_0 > 1$ there are oscillations due to reflections at the discontinuity of the potential.
- $T = 1$ for $k_2' a = \pi, 2\pi, \ldots$ a is equal to integral or half-integral of $\frac{2\pi}{k_2'}$

Examples of Barrier Penetration

1) Joining of copper wires. There is a layer of copper oxide which is a very effective insulator but the electrons flowing through the wire are able to tunnel through the layer.

2) Emission of α-particles by radioactive nuclei.

Rutherford had studied the scattering of α-particles by U^{238} nuclei. He used 8.8 MeV α-particles and the results were in agreement with his theory

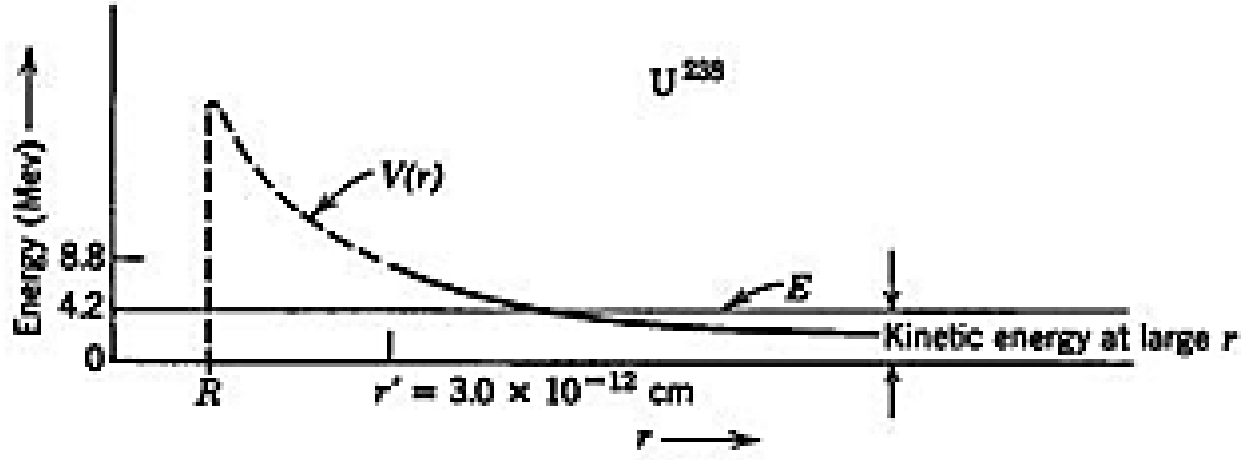
- The α-particle near a U^{238} nucleus felt a repulsive potential given by Coulomb's law $V = 2Ze^2/4\pi\epsilon_0 r$ at least for $r > r' = 3\ 10^{-14}$ m where $V(r') = 8.8$ MeV.
- Scattering from nuclei of light atoms showed that $V(r)$ departs from Coulomb's law for distances $r < R$, where R is the nuclear radius.
- α-particles are emitted by U^{238} nuclei which means that they exist inside the nuclei

Examples of Barrier Penetration

The potential $V(r)$ up to R must be as depicted in the Figure.

Paradox. The U^{238} nucleus emits α-particles with E = 4.2 MeV

An α-particle is initially in $r < R$ and this region is separated from the rest of space by a potential barrier such that $V_0 > 8.8$ MeV. However, it was observed that sometimes an α-particle penetrates the barrier and escapes.



igure 8–13. An energy diagram showing the relation between potential and total nergies in alpha particle emission.

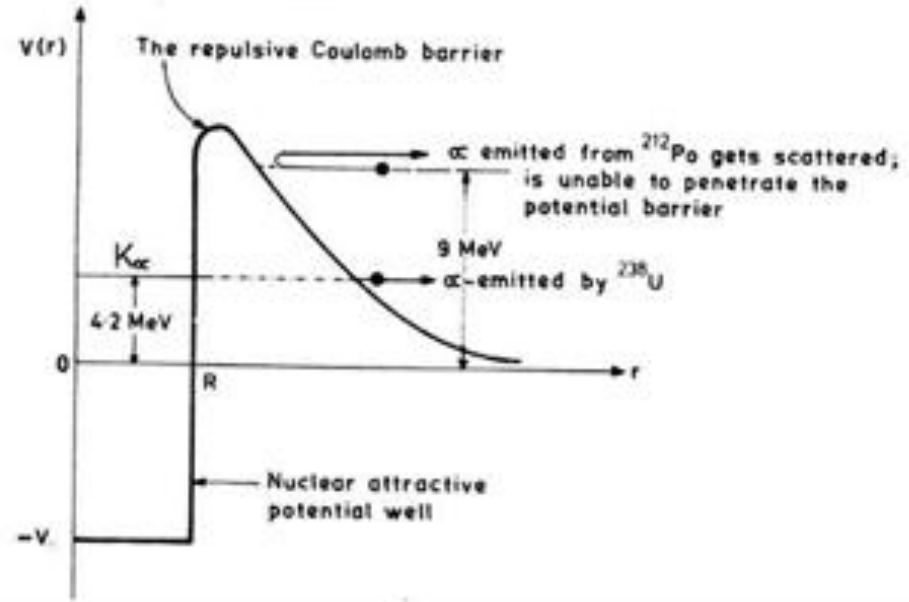
Examples of Barrier Penetration

Consider,

$$^{238}_{92}U \rightarrow ^{234}_{90}Th + \alpha \qquad KE(\alpha) = 4.2\,MeV$$

A 4.2 MeV α particle is able to come out of the Uranium nucleus

However, α particle with KE(α) =8.8 MeV (from ^{212}Po) is unable to penetrate $^{238}U_{92}$!



If 9 MeV α particle is not able to penetrate the Coulomb barrier from outside, then how is the 4.2 MeV α particle able to penetrate from inside ?

Examples of Barrier Penetration

Theoretical explanation (1928) Gamow, Condon and Gurney (GCG)

They considered the α-particle emission as a quantum barrier penetration problem. Assumptions:

- $V(r) = 2Ze^2/4\pi\epsilon_0 r$ for $r > R$
- $V(r) < E$ for $r < R$
- Use $T \approx 16(E/V_0)(1 - E/V_0)e^{-2k_2 a} \approx e^{-2\sqrt{(2m/\hbar^2)(V_0 - E)}a}$

The barrier $V(r)$ in the figure is considered as a set of adjacent rectangular barriers of height $V(r_i)$ and width Δr_i.

Now, the transmission for the i-th barrier is $T_i \approx e^{-2\sqrt{(2m/\hbar^2)(V(r_i) - E)}\Delta r_i}$

$$T \approx T_1 \cdot T_2 \cdot T_3 \ldots \approx e^{-2\int_R^{r''} \sqrt{(2m/\hbar^2)(V(r) - E)}dr}$$

where $V(r'') = 4.2$ MeV

Examples of Barrier Penetration

T gives the probability that in one trial the α-particle will penetrate the barrier AND the number of trials per second is $\approx v/2R$

The probability per second that the nucleus emits an α-particle (decay rate, DR) is

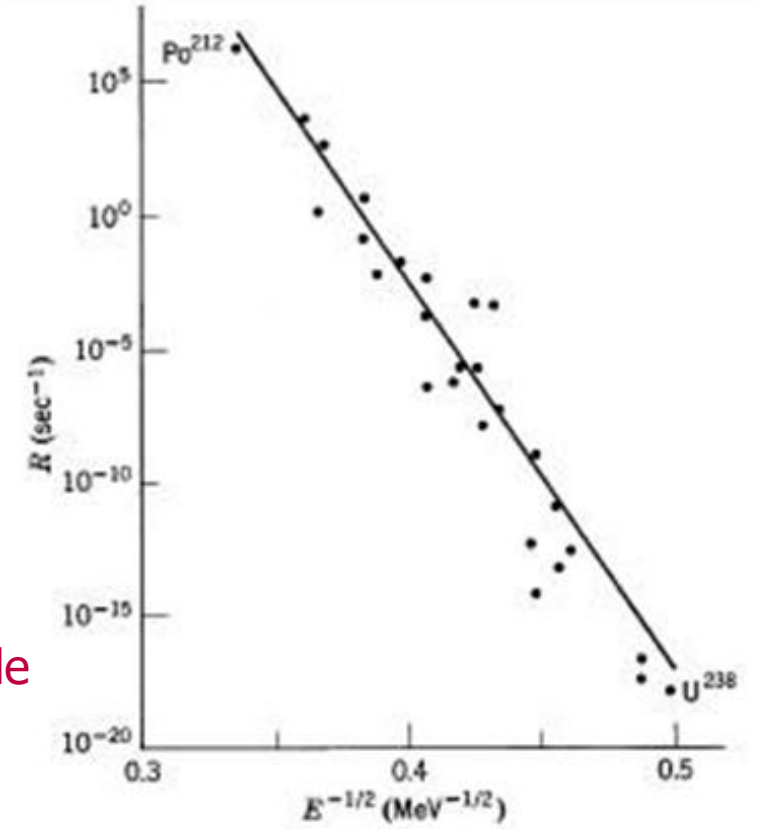
$$DR = \frac{v}{2R} e^{-2\int_R^{r''} \sqrt{(2m/\hbar^2)(2Ze^2/4\pi\epsilon_0 r - E)}\,dr}$$

All quantities were known, excepting *v* and R. GCG took *v* as the velocity of the α-particle after emission and R= 9 fm.

The calculated values were in good agreement with experimental data.

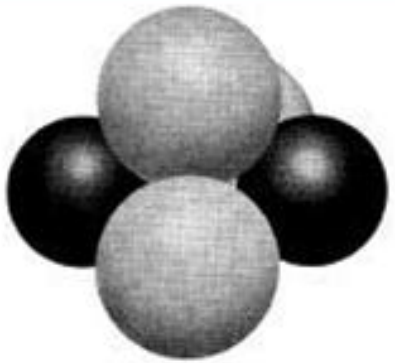
Notice that the decay rate (DR) varies over a very large range. For example

$DR(U^{238}) = 5 \cdot 10^{-18}\ sec^{-1}$ $DR(Po^{212}) = 2 \cdot 10^{6}\ sec^{-1}$

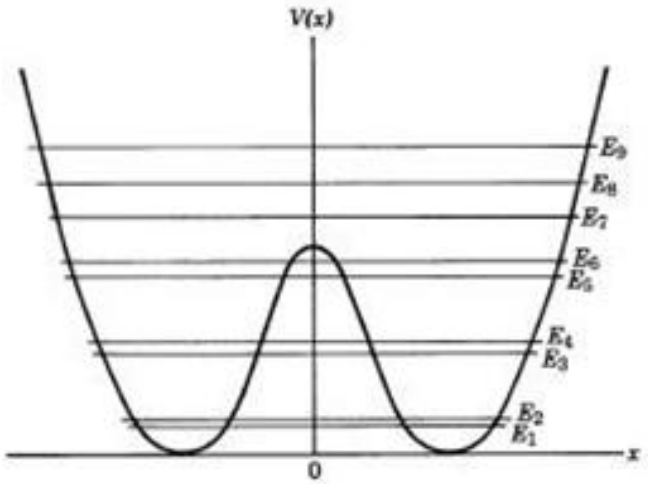


The probability per second *R* that a radioactive nucleus will emit an α particle of energy *E*. The points are experimental

Examples of Barrier Penetration



A schematic illustration of the NH_3 molecule. The light spheres represent the three H atoms arranged in a plane. The dark spheres represent two equivalent equilibrium positions of the single N atom.



The potential energy of the N atom in the NH_3 molecule, as a function of its distance from the plane containing the three H atoms, which lies at $x = 0$. In its lower energy states, the total energy of the molecule lies below the top of the barrier separating the two minima, as indicated by the eigenvalues of the potential shown in the figure.

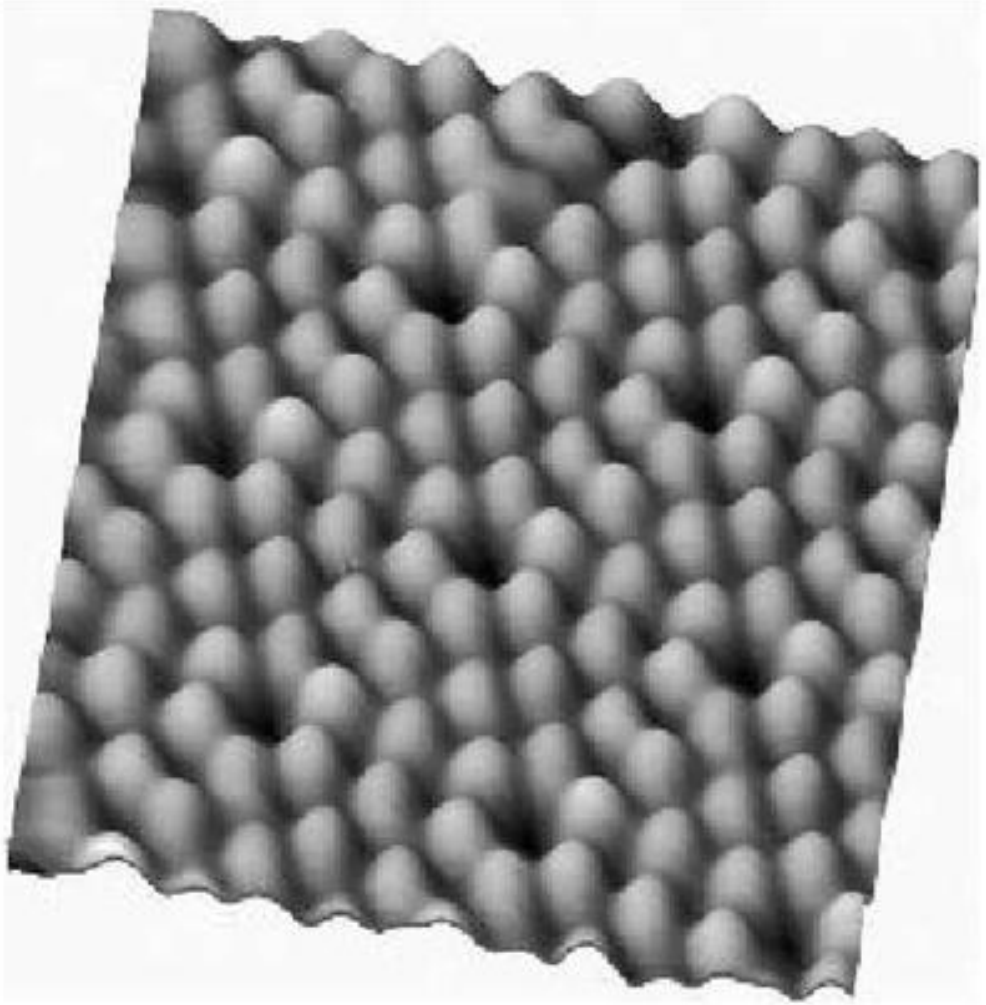
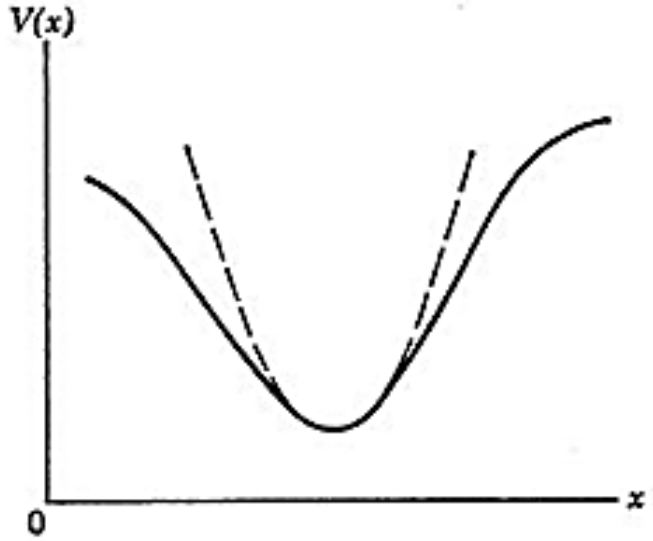


Figure 2.6. An image of the (111) surface of silicon obtained by scanning tunnelling microscopy. The bright peaks correspond to silicon atoms. The hexagonal symmetry is a characteristic feature of this surface. (Supplied by P. A. Sloan and R. E. Palmer of the

The 1-D Simple Harmonic Oscillator Potential

Very important. It is the prototype for any system involving oscillations.

- Vibration of atoms in diatomic molecules
- Atomic vibrations in solids (acoustic and thermal properties)
- Magnetic properties in solids (arising from vibrations in the orientation of nuclei)
- Any system with small vibrations about a point of equilibrium



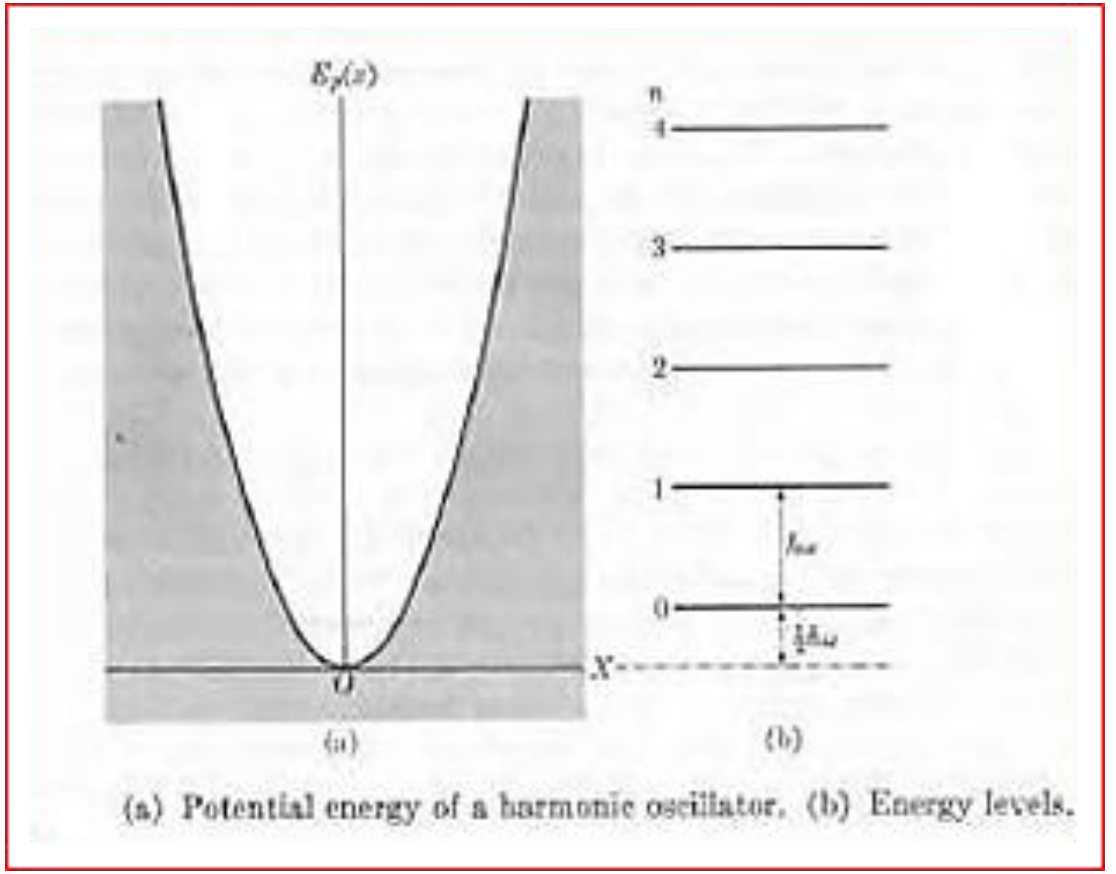
Illustrating the fact that any continuous potential with a minimum (solid curve) can be approximated near the minimum very well by a parabolic potential (dashed curve).

The 1-D Simple Harmonic Oscillator Potential

$$V(x) = K\frac{x^2}{2}$$

- **CM.** Simple harmonic motion with frequency $\nu = \frac{1}{2\pi}\sqrt{\frac{K}{m}}$
- The total energy E can take any value (is proportional to the amplitude).

- **QM.** Total energy is quantized
- $E_n = \left(n + \frac{1}{2}\right)\hbar\omega \qquad n = 0,1,2, \ldots$
- First eigenvalue $E_0 = \frac{\hbar\omega}{2}$ (zero-point energy). The particle cannot have zero energy.



(a) Potential energy of a harmonic oscillator. (b) Energy levels.

The 1-D Simple Harmonic Oscillator Potential

(A.I.M. Rae, *Quantum Mechanics*, 2002)

The potential is given as $V(x) = K\frac{x^2}{2}$ and we write $\omega = (K/m)^{1/2}$

We have to solve

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dx^2} + \frac{1}{2}m\omega^2x^2u(x) = Eu(x)$$

change variables $x \longrightarrow y$ where $y = (m\omega/\hbar)^{1/2}x$ define a constant $\alpha = (2E/\hbar\omega)$

so the eqns looks like

$$\frac{d^2u}{dy^2} + (\alpha - y^2)u(y) = 0$$

The 1-D Simple Harmonic Oscillator Potential

Now, $u(y)$ must remain finite when $|y| \to \infty$

Consider the asymptotic form of the solution

$$\frac{d^2u}{dy^2} - y^2u(y) = 0$$

General solution (in fact, it is of $\frac{d^2u}{dy^2} + (1 - y^2)u(y) = 0$)

$$u(y) = Ae^{-y^2/2} + Be^{y^2/2}$$

Finiteness implies B=0

Therefore

$$u(y) = Ae^{-y^2/2}$$

is the asymptotic form of the solution

The 1-D Simple Harmonic Oscillator Potential

Now, we try a solution (for all *y* values)

$$u(y) = H(y)e^{-y^2/2}$$

H(*y*) a function to be determined (but slow varying when $|y| \to \infty$).

Substituting, we end up with

$$H''(y) - 2yH'(y) + (\alpha - 1)H(y) = 0$$

We write (most general technique. Power series expansion)

$$H(y) = \sum_{p=0} a_p \, y^p$$

By evaluating the derivatives

$$H'(y) = \sum_{p=1} a_p \, p \, y^{p-1} \qquad H''(y) = \sum_{p=2} a_p \, p \, (p-1) \, y^{p-2}$$

The 1-D Simple Harmonic Oscillator Potential

We finally obtain

$$\sum_{p=0}\left[(p+1)(p+2)a_{p+2}-(2p+1-\alpha)a_p\right]y^p=0$$

Each coefficient must vanish.

Leads to the recursion relation

$$a_{p+2}/a_p=(2p+1-\alpha)/[(p+1)(p+2)] \qquad a_{p+2}/a_p=2/p \quad \text{when} \quad p\to\infty$$

We can write

$$H(y)=a_0\left(1+\frac{a_2}{a_0}y^2+\frac{a_4}{a_2}\frac{a_2}{a_0}y^4+\frac{a_6}{a_4}\frac{a_4}{a_2}\frac{a_2}{a_0}y^6+\cdots\right)+a_1\left(y+\frac{a_3}{a_1}y^3+\frac{a_5}{a_3}\frac{a_3}{a_1}y^5+\frac{a_7}{a_5}\frac{a_5}{a_3}\frac{a_3}{a_1}y^7+\cdots\right)$$

The 1-D Simple Harmonic Oscillator Potential

Both series have an infinite number of terms. Analyze them.

Compare to

$$e^{z^2} = 1 + z^2 + \frac{z^4}{2!} + \frac{z^6}{3!} + \cdots + \frac{z^l}{(l/2)!} + \frac{z^{l+2}}{(l/2+1)!} + \cdots$$

for which

$$a_{l+2}/a_l = \frac{1}{(l/2+1)!} \Big/ \frac{1}{(l/2)!} = \frac{1}{l/2+1} \approx \frac{1}{l/2} = \frac{2}{l} \qquad \text{(when } l \to \infty\text{)}$$

which means that the even series behaves (when $y \to \infty$) like e^{y^2} AND the odd series behaves (when $y \to \infty$) like ye^{y^2}

Therefore $H(y) = a_0 K_0 e^{y^2} + a_1 K_1 y e^{y^2}$ $\quad |y| \to \infty$

There is a convergence problem !!!

And finally $u(y) = C_0 e^{y^2/2} + C_1 y e^{y^2/2}$ $\quad |y| \to \infty$

The 1-D Simple Harmonic Oscillator Potential

Solution? Terminate the series after a certain number of terms. This can be achieved for **certain** values of α

$$\alpha = 2n + 1 \qquad \text{AND} \qquad \begin{cases} n = 1,3,5,\ldots & \text{if } a_0 = 0 \\ n = 0,2,4,\ldots & \text{if } a_1 = 0 \end{cases}$$

For example, generate the series with even powers. (it means $a_1 = 0$)

Remember that $a_{p+2}/a_p = (2p + 1 - \alpha)/[(p + 1)(p + 2)]$

$a_2/a_0 = (1 - \alpha)/(1 \cdot 2)$ if $\alpha = 1$ then $a_2 = a_4 = \cdots = 0$

$a_4/a_2 = (2 \cdot 2 + 1 - \alpha)/(3 \cdot 4)$ if $\alpha = 5$ then $a_4 = a_6 = \cdots = 0$

$a_6/a_4 = (2 \cdot 4 + 1 - \alpha)/(5 \cdot 6)$ if $\alpha = 9$ then $a_6 = a_8 = \cdots = 0$

and so on.....

The 1-D Simple Harmonic Oscillator Potential

The resulting solutions are the $H_n(y)$ polynomials, which are of order y^n and are called Hermite polynomials.

$$H_0(y) = 1$$
$$H_1(y) = 2y$$
$$H_2(y) = 4y^2 - 2$$
$$H_3(y) = 8y^3 - 12y$$

The eigenfunctions are

$$u_n(y) = A_n H_n(y) e^{-y^2/2}$$

As for the energies... (remember that $\alpha = 2E/\hbar\omega$)

From the condition $\alpha = 2n + 1$ we finally obtain

$$E_n = (n + 1/2)\hbar\omega \qquad n = 0,1,2,3,4, \ldots$$

The 1-D Simple Harmonic Oscillator Potential

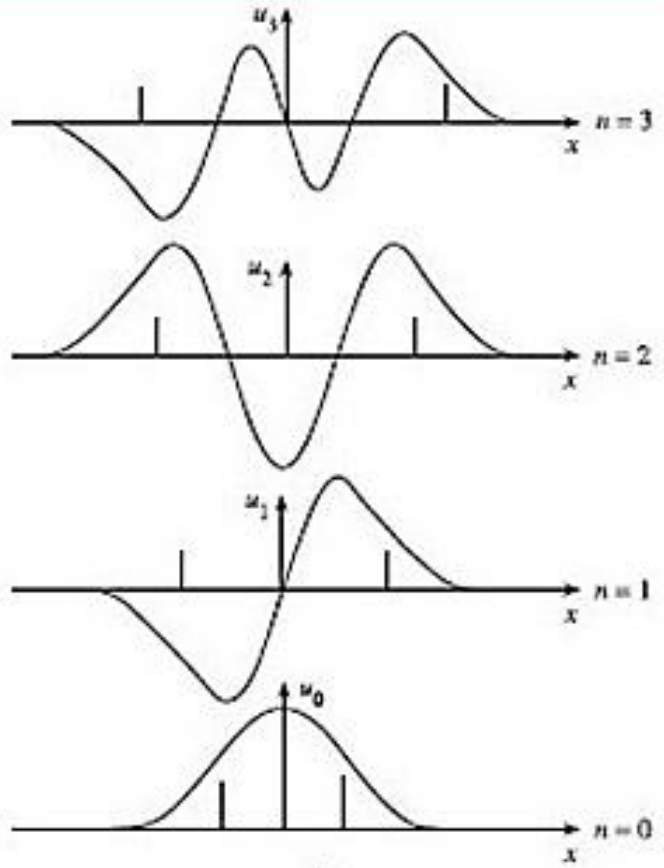
The polynomials $H_n(y)$ are known as Hermite polynomials.

$$\left.\begin{aligned} H_0 &= 1 \\ H_1 &= 2y \\ H_2 &= 4y^2 - 2 \\ H_3 &= 8y^3 - 12y \end{aligned}\right\}$$

where the values of a_0 and a_1 have been chosen according to established convention. The wavefunctions $u_n(x)$ can now be obtained by multiplying $H_n(y)$ by the factor $\exp(-\frac{1}{2}y^2)$, making the substitution $y = (m\omega/\hbar)^{1/2}x$

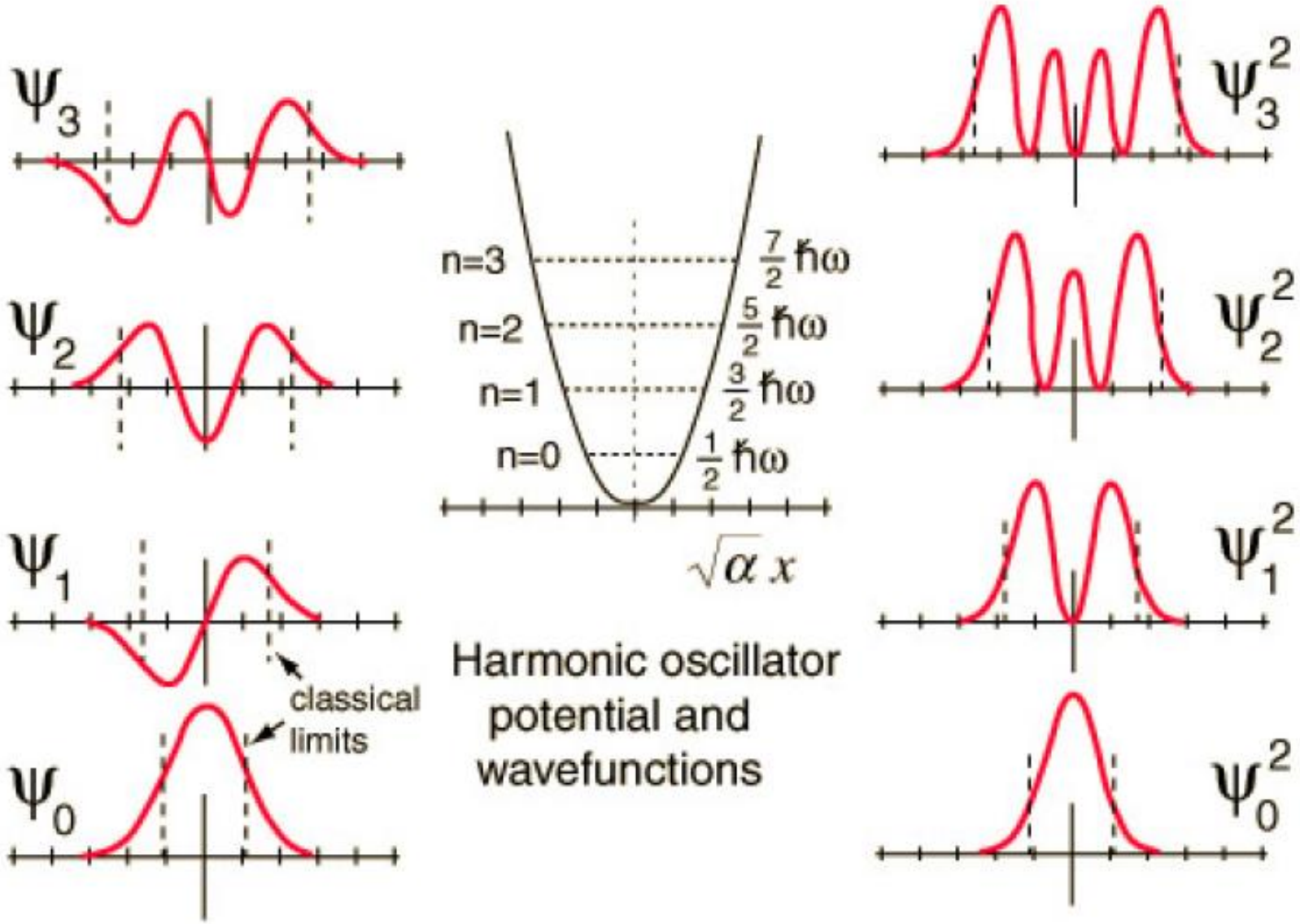
the wavefunctions are seen to have even or odd parity, depending on whether n is even or odd. We note that, like the wavefunctions associated with the energy states of a finite square well they are oscillatory within the classically permitted region and penetrate the classically forbidden region to some extent. Algebraic expressions for these wavefunctions are as follows:

$$\left.\begin{aligned} u_0 &= (m\omega/\pi\hbar)^{1/4}\exp(-m\omega x^2/2\hbar) \\ u_1 &= (4/\pi)^{1/4}(m\omega/\hbar)^{3/4}x\exp(-m\omega x^2/2\hbar) \\ u_2 &= (m\omega/4\pi\hbar)^{1/4}[2(m\omega/\hbar)x^2 - 1]\exp(-m\omega x^2/2\hbar) \\ u_3 &= (1/9\pi)^{1/4}(m\omega/\hbar)^{3/4}[(2m\omega/\hbar)x^2 - 3]x\exp(-m\omega x^2/2\hbar) \end{aligned}\right\}$$



. The wavefunctions corresponding to the four lowest energy states of t harmonic oscillator are shown in (*a*) v

The 1-D Simple Harmonic Oscillator Potential



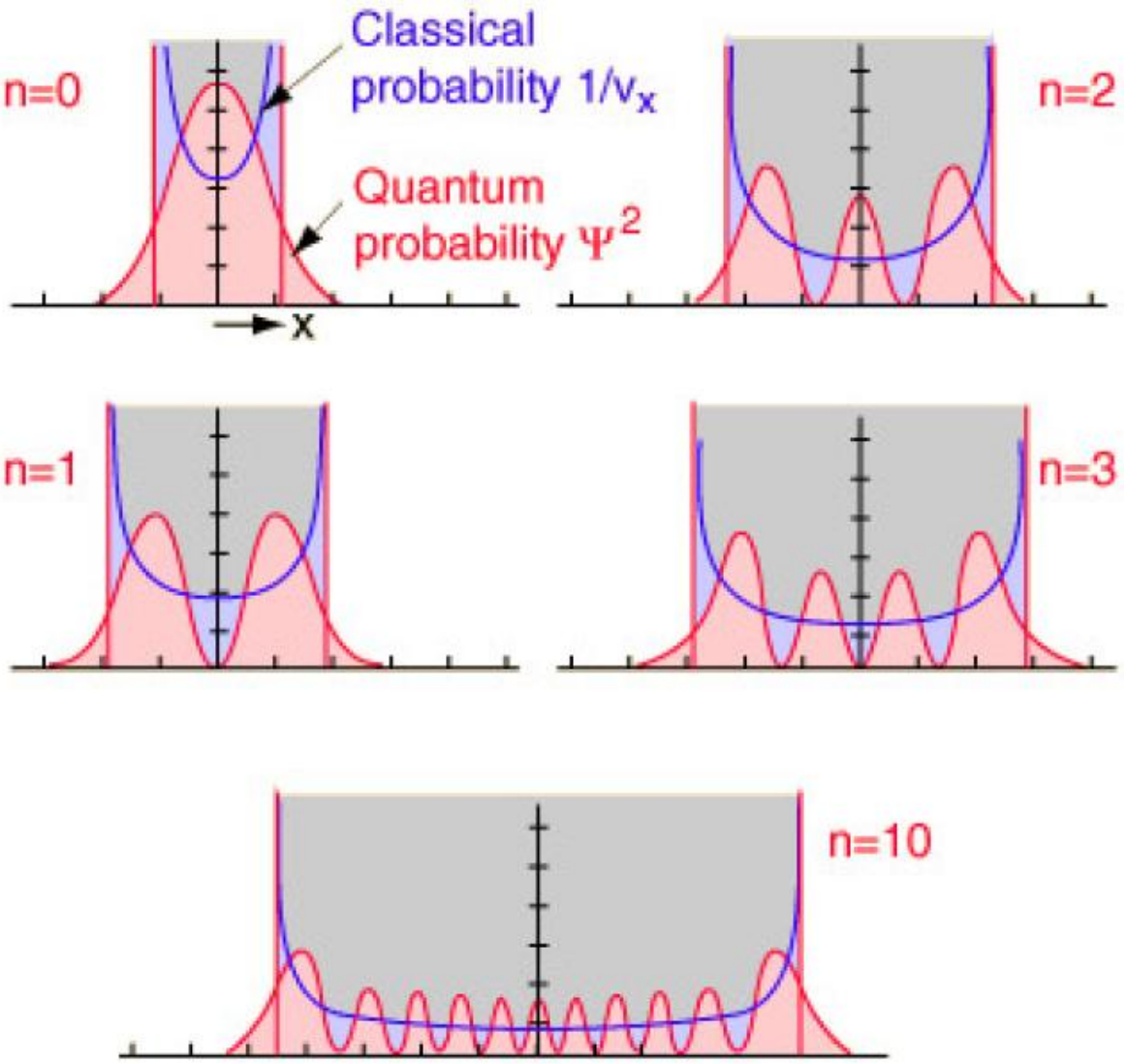
The 1-D Simple Harmonic Oscillator Potential

Classically

$$x = a\cos\omega_c t \qquad v = -a\omega_c \sin\omega_c t$$

where a is the amplitude of oscillation. Now an oscillating particle is more likely to be found near the limits of its oscillation where it is moving slowly than near the centre where it moves more quickly
the probability $P(x)dx$ that the particle be found in the region between x and dx is inversely proportional to the speed $v(x)$ at the point x.

$$P(x) = \frac{1}{\pi(a^2 - x^2)^{1/2}} \qquad |x| < a$$
$$= 0 \qquad |x| > a$$



The 1-D Simple Harmonic Oscillator Potential

$$V(x) = \frac{1}{2}Cx^2$$

Solution:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \qquad \omega = \sqrt{C/m}$$

$$\psi_n(x) = A_n e^{-\frac{\alpha^2 x^2}{2}} H_n(\alpha x) \qquad \alpha^2 = \frac{m\omega}{\hbar}$$

$H_n(u)$ are the Hermite's polynomials $\qquad n = 0,1,2,\ldots$

$$H_0(u) = 1 \qquad H_2(u) = 2 - 4u^2$$

$$H_1(u) = 2u \qquad H_3(u) = 12u - 8u^3$$